

Spare capacity, narrow surface: what a production agent-memory system actually surfaces

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Correction: the title changed on 2026-08-29, and the reason belongs to the paper itself. It used to be “*Spare capacity, starved coverage*”, and adversarial review pointed out two flaws in the metaphor. First, the coverage channel is not starved: it exhausts the eligible pool at 100% on every measured day (inside the defective-ingestion regime, §4.3.1). It is the corpus that is starved, not the coverage, so the image inverted the mechanism described in the body. Second, *starved* is normative, and §4.5 explicitly refuses that judgment: “a reader who concludes ‘the system is losing valuable information’ has gone beyond what was measured”. A subtitle cannot sell the conclusion that the body declines.

Status (2026-10-05). v1.0 of this manuscript (Portuguese) was published on Zenodo on 2026-08-30 (10.5281/zenodo.22181415). This is v1.1: the text revised after the 2026-09-21 corrections and the 2026-10-04 audit of aged claims (Appendix F), translated to English, and deposited on 2026-10-05 as v1.1 of that record, DOI 10.5281/zenodo.23163119 (version 2 of concept 10.5281/zenodo.22181414). Rule of this file: where there is a number, it comes from an artifact locked by `--assert-json` and its reproduction command is cited, except the numbers the text itself flags as having no preserved artifact: the 7,908 (81%) of §4.3.2; the 5.6× partial-day reading and the 1,971, 10,926 and 20.5 days of §4.3.1; the strata counts of §5.7 (1,139 / 951 / 311 / 57 and 14 / 61 / 89 / 186); the 19 of 32 and the 591,323 of §6.1; the pre-cleanup warning density (87 markers, 77 of 288 paragraphs, 26.7%) and the per-section densities of Appendix F; the survey counts of §1 and the 46.9% of §6 (the last two marked “—” in Appendix D).

Sources: SUPERFICIE-2026-08-27.md · REPLAY-OPORTUNIDADE-2026-08-27.md · REMEDIATION-2026-08-27.md · DEVIATIONS-FOR-PAPER.md · PROTOCOL-CALIBRATION-2026-08-27.md.

Abstract

Memory systems for agents are evaluated by retrieval quality over sets of queries. Among the 218 papers of the field’s canonical survey, none measures what the agent receives in production. Retrieval is also conditional on a query having been issued, so an item that no query reaches has an undefined nDCG, not a low one. For 12 weeks we instrumented the two surfaces through which a memory system in operation delivers content to a fleet of 6 agents: a proactive 10-item brief and on-demand search.

The brief delivered 583,763 slots, 8.7 times the size of the corpus, enough to serve each of the 67,187 chunks eight times. It delivered 1,635 distinct live chunks: 2.43% (1,787 counting the 152 that were served and later deleted; see §4.1). The aggregate capacity, therefore, did not force this result: there was room to show the whole corpus eight times over. It does not follow that the ordering is wrong, only that the number was not produced by a lack of space. The per-session capacity (10 items) is not tested here. (The 99.98% expected coverage under uniform random service, against a 100% maximum, illustrates what the capacity would allow; it is not a recommended policy.) Adding search, 56,288 live chunks (83.78%) have neither a brief-log record nor a positive search counter. This is a lower bound on non-delivery by the brief and by tracked search; non-delivery across all agent-facing search is not established (§3.1). The change of universe between the two

sentences is deliberate and must be read as such: the capacity slack above is the brief’s, whereas the 83.78% counts the union. Of the 10,899 live chunks with a record on either surface, 9,755 have a positive search counter. That counter records that a chunk was a top candidate of some tracked search call, whoever initiated the call (automated callers included) and whether or not the chunk was then returned: it bounds from above what tracked search returned, and it shows neither who asked nor how much search actually delivered (§3.1). The brief, the surface that proactively selects content for delivery, exposed 1,635 live chunks (1,787 counting the 152 later deleted). Every claim in this paper about *mechanism* concerns the brief; the 83.78% describes the state and does not assign a cause.

The two channels of the surface freeze, for opposite reasons and neither tied to capacity. The 8 slots of the main pool are ordered by a score whose terms, with one exception, do not decay: the access component is monotone in a counter that only goes up. The 3 chunks present in 100% of the 4,632 briefs of the week occupy positions 1, 3 and 4 (last accessed 42, 90 and 30 days before the close of the window, 2026-08-28; §4.3.2). Tracked search traffic from months ago, whoever initiated it (§3.1), determined those positions, and the top-10 takes 47.16% of the slots. The other 2 slots are a *coverage* channel, whose declared purpose is to serve the never-served, and it freezes for a different reason: its eligible population is 108 chunks in a corpus of 67,187 (0.16%, carved out by two path patterns together with the channel’s importance floor and 30-day window), and it exhausts that population entirely on every measured day, with 12.4 slots per candidate on a closed day (an unpreserved reading on a partial day gave 5.6, and the two are not comparable; see §4.3.1). On every measured day there was no never-served item left to serve. These coverage numbers were measured on 2026-08-26 to 08-29, inside a period of defective session ingestion (§4.3.1), and describe the channel in that regime, not its steady state. The main pool would respond to a score adjustment, and nobody adjusts it. The coverage channel, designed to compensate for the main pool, has a daily reach bounded by its eligible population (path patterns, the importance/pain floor and age windows), which it exhausted in that regime; within a brief it responds to score only within `last_served` ties (17/350 here) and then saturates.

The mechanism of the coverage channel is deducible from the code. It orders by a lexicographic comparator (`last_served ASC, salience DESC`): the score is the subordinate coordinate and decides only within ties of the dominant one. That predicts a ceiling, not a proportional response, for any additive bonus on the score of that channel. We tested with an increasing dose by counterfactual replay over 350 of 350 real brief states, faithful to the serving pipeline. Caveat: the states are from production; the intervention was not served. The mode is *shadow*: the treated composition is computed and recorded, and what the agent received during the measured window was always the control (§7). Result: a monotone response in each state, saturation at $w \in (4.0, 4.4]$, and a ceiling of 4.86% of briefs (measured on the 2026-08-26 corpus, inside the defective-ingestion regime of §4.3.1, with an empty never-served stratum). The ceiling is also not a constant of the mechanism, along two axes we measured: under the same rule with another draw of designated items it reaches 7.43% (the draw in force sits at the minimum of the distribution, tied with another), and truncating the timestamp resolution from second to minute or hour takes it to 36% and 80%, without changing a single line of code. The reach of the mechanism is fixed by decisions that nobody took as policy. Correction: a third axis exists, and an audit on 2026-08-30 found it in an artifact we had recorded and never read: excluding the rows of our own health probes moves the replay’s anchor. The ceiling above was computed without excluding them. Measured the same day with a paired design, excluding them lowers it from 17/350 (4.86%) to 13/350 (3.71%), and only one sensitive state is common to the two arms (§5.7.2).

Caveat: what we do not claim. No effect on agent behavior: no downstream outcome is instrumented (§5.4). Nor do we claim that the concentration is *wrong*. A policy that serves 10 items per session should concentrate, and serving uniformly would be useless; the finding is that the non-exposure is a result of policy and not a capacity limit, hence revisable by a design decision. And we claim nothing about the field: this is one system. The generalization of the mechanism is deductive and holds for any ranker with lexicographic order and a bonus on the subordinate coordinate that serves a prefix of that order. How many systems have this shape is an open question, and the executable diagnostic we publish exists so that others can answer it one at a time.

Caveat: two notes on reading. First, of the 10,899 chunks with an exposure record, 9,755 have a positive search counter, an upper bound on what tracked search returned (§3.1) that records candidacy in a tracked call, not who initiated it; the brief proactively selects content for delivery, and the mechanism claims hold for it. Second, collection size, which correlates with exposure (§4.2), may be a proxy for how the type is

produced: curation is not ruled out as a common cause.

1. Introduction

An agent memory system is judged today by retrieval quality: given a set of queries, how well it ranks what is relevant. This is the question the benchmarks answer and the one engineering optimizes (better embeddings, reranking, query expansion). It presupposes, without saying so, that what the agent receives is the top of that ranking.

A question comes before it and goes unasked: what does the agent receive? It cannot be answered with a set of queries, because it requires the system in operation. It can be answered, because every delivery passes through a small number of surfaces that can be instrumented. Here there are two: a proactive 10-item brief at the start of each session, and on-demand search.

The expected answer would be “it doesn’t fit”, and the measurement says it fits. In 84.7 days the brief delivered 583,763 slots against 67,187 chunks, enough capacity to serve every chunk 8.7 times. It served 1,635 distinct live chunks, 2.43% of the corpus (1,787 in the historical count, which includes 152 deleted afterwards; §4.1 gives the reason both numbers exist). Under uniform serving the expected coverage would be 99.98%. Adding search, 83.78% of the corpus has no exposure record on either surface, a lower bound on non-delivery by the brief and by tracked search (§3.1). The number of slots does not impose the non-exposure. Caveat: the part that the ranking explains is the brief’s, and §4.1.1 delimits what can be attributed to each surface. The claim about mechanism is about the brief, and the aggregate number measures what has no record of arriving, not what the ranker refused.

Caveat: this is not an accusation against the policy. A 10-item surface must concentrate; serving memory at random would be worse than not serving it. What the number changes is the nature of the problem. As long as the surface is believed to be too small, non-exposure is a fact of life; once it is measured to be $8.7\times$ the corpus, non-exposure becomes a policy choice. There are in fact two choices, one per channel: the coverage channel exhausted a pool of 108 chunks selected by path patterns, the importance/pain floor and age windows in the measured regime, and the main pool serves 33 distinct chunks on every measurable day but one (the identical set from 2026-08-24 to 2026-09-19) from 5,376 main slots on a 672-brief day because its ranking is a deterministic top-k with no serve-history term (§4.3.1). Choices can be examined.

We examined them, and we located the concentration. The 8 slots of the main pool converge: 3 chunks appear in 100% of briefs, placed there by the high-pain floor ($\text{pain} \geq 0.9$; measured over 2026-08-21 to 09-21, §4.3.1), which takes all three corpus-wide slots, and the top-10 takes 47.16% of a week’s slots. The remaining 2 slots are a coverage channel, which exists precisely to serve the never-served. That channel has two distinct constraints: eligibility (path patterns, the importance/pain floor and age windows) bounds daily reach; holding eligibility fixed, an additive salience bonus changes per-brief selection only within `last_served` ties:

1. **Calendar:** the channel cuts by age, and ingestion arrives in batches. Between batches the pool is empty and the channel serves the same set for days: we measured five consecutive days with zero new items, with the minimum age of what was served rising by exactly +1.00 per day, the signature of a frozen set. We later established that those days fell inside an ingestion defect, not the normal cadence of ingestion: session ingestion produced zero chunks from 2026-08-11 until its repair on 2026-09-07 (DEVIATIONS-FOR-PAPER.md §10.7; §4.3.1), so they describe the channel under defective ingestion, not its steady state. The window is not unique: there are two sub-pools, the per-agent one with 7 days and the global one with 30 (§4.3.1), and the five-day observation belongs to the first. Applying the per-agent sub-pool’s window to a batch of the global sub-pool is an error we made and that a registered prediction refuted (F-2);
2. **Algebra:** the channel orders by a lexicographic comparator (`last_served` ASC, `salience` DESC), in which the score is the subordinate coordinate and decides only within ties of the dominant one. Deductively, from seven lines of code, this predicts a ceiling for any additive bonus on the score of that channel.

The prediction is testable and we tested it, with increasing dose in counterfactual replay over 350 of 350 real brief states, faithful to the serving pipeline. Caveat: these are production states, and the intervention was not served (*shadow* mode in the window this paper measures, §7). The response is monotonic in each state and saturates at $w \in (4.0, 4.4]$, with a ceiling of 4.86% of briefs (measured on the 2026-08-26 corpus, inside the defective-ingestion regime of §4.3.1, with an empty never-served stratum). The prediction survived the test that could have killed it, as well as an earlier instrument that confirmed it for the wrong reason (§5.6).

Caveat on scope. It is one system. We did not measure the effect on the agent’s behavior; no downstream outcome is instrumented (§5.4). We do not claim that the field optimizes the wrong coordinate. We claim that there is a coordinate the benchmarks do not measure, we give the instrument to measure it, and we leave the question open. Of the two surfaces, the brief proactively selects content for delivery; the other, search, is instrumented by a counter of candidacy in tracked calls, which records neither who initiated a call nor what it returned, so its share of delivered exposure is not established (§3.1, §4.1.1).

- **The gap:** the field’s canonical survey (TMLR 2602.06052v4, 218 papers) maps agent memory architectures and benchmarks. Benchmarks measure nDCG/recall over sets of queries. None measures the delivery surface: how many distinct items an agent in production sees, and which ones.

The vocabulary of experimental methodology is also absent. Recomputed over the v4 PDF (`measurement/survey-string-count.py`, sha256 497e9549...b46a6, 429,387 characters, 63 end-of-line hyphenations stitched together before counting):

term	body	bibliography
pre-registration / preregistration (and the 6 other spellings)	0	0
randomized / randomised	0	0
ablation / ablations	0	0
interventional	1	0
counterfactual	1	0

The absence covers the whole family of terms, not one word. A survey of 218 papers that says **memory** 1,208 times (the positive control’s count over the whole extracted text, bibliography included; the script emits no body-only count for **memory**) and **randomized** never describes a field whose instrument is the offline benchmark, not the experiment. The two occurrences that exist are singular, and one of them, that of **counterfactual**, appears as a suggested future direction.

Caveat: zero is the result that a broken extraction produces for free, so the count runs with a positive control (**memory**, **agent**, **benchmark**, **evaluation** above floors) and aborts if it fails. The control did fire once, on **ablation**=0. The control’s floor for **ablation** was what was wrong (a survey catalogs, it does not ablate), and the direct check (**memory**=1,208, **benchmark**=126 in the same text) showed the extraction was intact. The term left the control and became data.

- **Why the question matters:** ~~If the surface has fixed, small capacity, then improving ranking does not improve exposure, and the field optimizes the wrong coordinate.~~ That was the hypothesis this work started with, and the measurement contradicts it: the surface is not small; it is $8.7\times$ the corpus. What matters is what remains after that: a surface with slack delivers 2.43%, and the channel that would exist to compensate for it is governed by an eligibility predicate (two path patterns, an importance floor and an age window) that admits 0.16% of the corpus, and by a lexicographic order in which the score decides only within ties of the dominant coordinate. Whether other systems have this shape is an open question, not a claim of this paper, and the published diagnostic exists so that it can be answered.
- **Contributions:** (i) the measurement of the exposure surface of an agent memory system in production, with the result that capacity exceeds the corpus by $8.7\times$ and yet 83.78% has no exposure record on either surface, a lower bound on non-delivery by the brief and by tracked search (caveat: that number sums the two surfaces, whereas the capacity cited is only the brief’s, and §4.1.1 delimits what each one authorizes concluding); (ii) the localization of the bottleneck in the coverage channel, with the two mechanisms that freeze it: an eligible population of 108 chunks (0.16% of the corpus), cut by

path patterns, an importance floor and age windows, and measured inside a period of defective session ingestion (§4.3.1), and a lexicographic order that demotes the score to a subordinate coordinate; (iii) a deductive ceiling prediction for additive bonuses in that channel, tested with dose-response and faithful replay; (iv) the executable diagnostic (`measurement/`), so that the measurement is reproducible on another system.

Caveat: the catalog of instrument defects (Appendix E) does not count as a contribution. It lists 17 defects that we committed, eight of them altering a number that this paper reports. We report them as an obligation, and eight findings do not bound the non-findings. They are in the appendix because whoever reproduces the measurement will fall into the same ones.

2. System under measurement

The system is the persistent memory of a fleet of 6 coding agents in continuous operation; during the measured week (2026-08-20 → 08-27) it served 4,632 briefs, ~660 per day (`out/superficie.json`, `briefs_7d`). The corpus is a single SQLite database with lexical search (FTS5), dense vectors and an entity graph; at the measurement instant (2026-08-28) it held 67,187 chunks. A chunk is a unit of text with a type (`lesson`, `decision`, `daily`, ...), a date of origin and three scalars that feed the ranking: `importance`, `pain` and `access_count`. None of this is specific to the system; what matters for the paper is the shape of the surface, not the implementation underneath.

There are exactly two surfaces through which a chunk can reach an agent, and this is what makes the population measurable:

1. the proactive brief (`/api/brief`), assembled at the start of each session with 10 items. Every organic brief in the §4.3 window has 10; the only exceptions are our own five 5-item health probes (§4.3; `RECON-52-e-sondas-2026-10-04.json`, `c25`). No agent asks for it: it receives it;
2. **on-demand search**, when the agent decides to look something up.

Every exposure goes through one of the two. “Never exposed” is therefore a verifiable property of the records, not an inference: it is the absence of a record in both. Read as non-delivery, it is a lower bound on non-delivery by the brief and by tracked search; non-delivery across all agent-facing search is not established (§3.1).

The mechanism lives in the composition of the brief. Of the 10 slots, 10 - `freshSlots` come from a main pool ordered by `salience` (an additive sum of importance, recency, pain and access), and up to `freshSlots` are reserved for a coverage pool, whose declared function is to serve what has not yet been served. The coverage pool is ordered by a lexicographic comparator (`last_served ASC`, `salience DESC`).

Three parameters of this description are themselves results, and §5 establishes them:

- `freshSlots` = 2 in production during the measured window, and by *configuration default with no override* (verified on 2026-08-27, §5.3): the number that governs the entire renewal of the surface was never chosen;
- it is a ceiling, not a quota: the 2 slots are filled *if* there is an eligible candidate. No measured day lacked one: §4.3.1 finds 108 eligible on each of 2026-08-26 to 08-29, and coverage filled exactly 2 slots in each of the 40 briefs attributed by channel (`CHANNEL-ATTRIBUTION-2026-08-29.json`). The five consecutive days that §1 cites are days on which no new item entered the pool (§4.3.1);
- and the comparator being lexicographic, rather than a weighted sum, is what gives `salience` the role of a subordinate coordinate inside the coverage channel.

Caveat. This is one system. The generalization of the mechanism (§5) is deductive, from the algebra of the comparator, and not empirical; how many systems share this shape is an open question, which the published diagnostic allows to be answered one system at a time.

Figure 0 (`out/fig0-arquitectura.svg`): corpus → two channels → 10 slots, with the comparator annotated on the coverage channel and its 2 slots highlighted.

Caveat. It is the only figure in the paper that does not derive from data: an architecture diagram is a claim about the code, not about a measurement, and the geometry is hand-drawn. But every number labeled in it comes from `out/superficie.json`, because a typed label silently ages into falsehood, and in this project one already has. The generator aborts if the identity `live-exposed + never-exposed = corpus` stops holding; when the generator was written, this check exposed that the naive sum exceeds the corpus by 152.

3. Method

3.1 The two surfaces, and what the count is exact about

surface	instrument	coverage
brief	<code>brief_log</code>	the endpoint’s entire lifetime (went live 2026-06-04) and no pruning — the only <code>DELETE FROM brief_log</code> in the repository is in a test
search	<code>chunks.access_count</code> , incremented for the top candidates of each tracked search sub-query	since always; the brief never writes to that column

Both instruments cover the period from the start, but they do not record the same event. `brief_log` records what the brief returned. `access_count` is incremented inside `search()` and `searchSemantic()` for each sub-search’s own top candidates (`search.ts:396`, called at `:466` and `:579`), and `searchHybrid` runs several such sub-searches (FTS on the original query and on each expansion variant at `2.limit` each, semantic at `4.limit`) before RRF fusion, truncation and `dedupe` cut the answer to `limit` (`search.ts:616-676`). A positive counter therefore marks a chunk that was a top candidate of at least one tracked sub-search: a superset of the chunks tracked search returned. Tracking is switched off per call by `trackAccess = false`, which the healthchecks and the semantic canary use (`search.ts:379-380`), so what untracked calls returned leaves no counter. Tracking is also the default, and the counter is cumulative: the same comment records that the semantic canary, an automated caller, incremented the counter of its top candidates on every cycle until it was switched to untracked (`search.ts:377-396`). A positive counter therefore records search candidacy in some tracked call; it records neither who initiated that call (an agent or an automated caller) nor whether the chunk was returned. The complement (live chunks with neither a `brief_log` row nor a positive counter) is an exact count of that predicate and a lower bound on the live chunks returned by neither the brief nor tracked search. Extending it to every search return would require showing that no agent-facing call ran untracked, which is not established here.

Correction. The union is not, and the earlier version of this sentence said it was (corrected 2026-09-21, by adversarial review). The two legs have different durabilities, and the table above shows why: `brief_log` is a durable, unpruned log, so it records exposure of chunks that have since been deleted; these are the 152 in §4.1. `chunks.access_count` is a column of the live row: a chunk exposed by search and deleted afterwards takes its counter with it, and the per-chunk telemetry that could have preserved it had no writer from 2026-05-19 14:47Z until its writer was restored on 2026-08-27 22:18Z (commit 32f78109; §6). Exposure by search followed by deletion within that interval is therefore invisible by construction.

Hence the historical union of 11,051 is not a bound in either direction on delivered exposure: the search leg loses chunks deleted after exposure (downward) and counts candidates that were never returned (upward). Only the live complement is exact, as a count of its predicate; read as non-delivery by the brief and by tracked search, the 56,288 (83.78%) is a lower bound, and non-delivery across all agent-facing search is not established. The asymmetry changes no number in this paper, but it changes what can be said about the union, which must not be reported as a count.

3.2 Measurement discipline

Five rules, each of which exists because violating it has already produced a wrong number in this work (§6):

1. **window closed at both ends**, with the `sha256` of the slice;
2. **derivation in script**, never in prose: prose that asserts a computed result is a cache with no invalidation. Every number in the paper comes from `measurement/*.py|mjs` and is locked by `--assert-json`, except those flagged where they occur as having no preserved artifact (listed in the status block at the top);
3. **reproduce a published anchor before varying anything**;
4. **exact vs. bound** declared per number, with the direction the bound protects;
5. **replay through the real code**, imported, never reimplemented.

3.3 Replay fidelity

The interventional evidence depends on a replay of the serving pipeline. It imports the composition function from the production build as of 2026-08-27/28 and is validated, brief by brief, against what production recorded. (Code line citations in this paper refer to `brief.ts`, `brief-diversity.ts`, `salience.ts` and `search.ts` as deposited in `serving-*.ts`; the salience formula of §4.3.2 is `calculateSalience` in `serving-salience.ts`; `brief-diversity.ts` is pinned by `sha256 34c9aee5....`)

serve-state cut	briefs	control matches	churn matches	production churn	replay churn	invented	lost
by insertion order	350	350	350	12	12	0	0
by timestamp (strict)	350	—	346	12	14	3	1

Checking the composition of the control arm prevents fidelity by coincidence: without that column, a brief whose control and treated arms were both wrong could still match on the outcome.

```
replay-resumo.py --campo out/c-350-v3.json --campo-estrito out/c-350.json
```

4. Results

4.1 Exposure: 83.78% of the corpus has no exposure record — and 74.75% of what passes the coverage channel’s importance floor

live corpus at 2026-08-28	67,187
exposed in the brief (historical)	1,787
positive search counter (live)	9,755
historical union	11,051
— of which, deleted afterwards	152
live union = 11,051 – 152	10,899
no record on either = 67,187 – 10,899	56,288 = 83.78%
of these, pass the coverage channel’s importance floor	10,008

The table closes over one universe: the live corpus. The bridge between the historical union (11,051) and the live complement is the row for the 152 chunks served in the brief and deleted afterwards: $11,051 - 152 = 10,899$ live exposed chunks, and $67,187 - 10,899 = 56,288$ ($\leftrightarrow$ F-3.2). The search row is a live count (`access_count > 0` on the rows that exist, `superficie-de-exposicao.py`), so the historical union mixes the historical brief with live search and is not a bound on delivered exposure (§3.1). The complement row is exact as a count of its predicate: 56,288 live chunks (83.78%) have neither a brief-log record nor a positive search counter. This is a lower bound on non-delivery by the brief and by tracked search; non-delivery across all agent-facing search is not established.

The importance-floor row belongs in the headline. A reader may grant the aggregate and reject the consequence: if most of the 56,288 were low-value noise, there would be nothing to deliver. That is not the case: of

the 13,388 chunks that pass the importance floor of the coverage channel, 10,008 = 74.75% have no exposure record.

This importance floor belongs to one channel, not to the system. `freshMinImp` and `freshMinPain` live in `DIVERSITY_DEFAULTS`, next to `freshSlots: 2`, and enter only through `fetchFreshCandidates`: they are the eligibility criterion of the 2-slot coverage channel. The 8 slots of the main pool apply no importance filter at all. (F-2)

Caveat: the two sides of the ratio also come from different universes, which constrains the reading: the numerator counts the absence of a record on both surfaces (`brief` $\cup$ tracked search), whereas the denominator is the criterion of one channel. The quantity is legitimate (“the coverage channel would consider this chunk, and neither the brief nor tracked search has a record of showing it”), but it is not “the system judged it relevant and refused to serve it”.

Caveat: the importance floor is also not a constant of the code. The *form* of the predicate is at `brief.ts:642`: `(COALESCE(importance,0) >= ? OR COALESCE(pain,0) >= ?)`, and the `OR` is the code’s, not ours. The *values* 0.7/0.7 are at `brief-diversity.ts:59–60`, and they are defaults that can be overridden at runtime by `NOX_BRIEF_DIV_FRESH_MIN_IMP` and `NOX_BRIEF_DIV_FRESH_MIN_PAIN (:88–89)`. Verified in the server process on 2026-08-29: neither is set, neither in the process environment nor in the `.env`, so the defaults apply and the numbers above are the production ones ($\leftrightarrow$ F-3.3).

Conditioning on declared relevance lowers the rate by 9 points and the absolute value by 5.6 $\times$, but does not dissolve it. The complement is the larger part: of the 56,288 with no record, 46,280 (82.2%) do not pass even this importance floor. Anyone who argues that non-exposure is, to a large extent, correct filtering of irrelevant material has those 82% on their side. What the co-headline establishes is that 10,008 chunks remain (14.9% of the corpus) that pass the coverage channel’s importance floor and that neither the brief nor tracked search has a record of showing.

Passing the importance floor is not being eligible, and the difference is two orders of magnitude. The channel applies three joint conditions: the importance floor (`importance  $\geq$  0.7 OR pain  $\geq$  0.7`), a path-pattern cut and an age window. The importance floor alone selects 13,388; the three together select 108 (§4.3.1), and of these, zero were never served. The 10,008 are not “chunks the channel wanted to serve and did not”: they are chunks whose importance the channel would recognize if they reached it, and what keeps them from reaching it is the eligibility predicate (path patterns and age windows; which of the two binds was not decomposed per chunk), not the ranking ($\leftrightarrow$ F-2).

Caveat: the tempting reading of this number is false. Of the 10,008, 8,928 (89.2%) are **distilled**: session fragments averaging 232 characters (`measurement/composicao-do-piso.py`, `out/FLOOR-COMPOSITION-2026-08-29.json`). The finding is not “ten thousand invisible lessons”; it concerns fragments that pass the coverage channel’s importance floor and have neither a brief-log record nor a positive search counter. Their count is a lower bound on non-delivery by the brief and tracked search; non-delivery across all agent-facing search is not established ($\leftrightarrow$ F-3.4; §3.1).

The rate is unconditional, and that is the objection whose direction is unknown. A chunk created in week 11 of the window had 7 days of opportunity for exposure; one from week 1 had 84. Both enter the denominator equally. Had the corpus grown quickly near the end, a slice of the 56,288 would be “too new to judge” rather than “not delivered”. Stratifying by age (`measurement/exposicao-por-coorte.py`, `out/EXPOSURE-BY-COHORT-2026-08-29.json`):

cohort	chunks	never exposed	%
< 1 week	96	90	93.75%
1–4 weeks	1,213	753	62.08%
4–12 weeks	4,553	3,013	66.18%
> 12 weeks	61,325	52,432	85.50%

The young cohorts do not drive the headline. The oldest cohort (91.3% of the corpus) is more unexposed than the aggregate (85.50% against 83.78%); only the < 1 week cohort ($n = 96$) is higher, and the two young

cohorts together are 1.95% of the corpus, too small to move the aggregate in either direction. This is a descriptive comparison, not a censoring correction: the cohorts differ in composition, age is measured from `COALESCE(source_date, created_at)`, which can precede ingestion, and additional follow-up could only move a young chunk from never exposed to exposed. We report the aggregate.

Caveat: a third leg of the same objection is uncountable by construction, and is declared: a chunk created and deleted within the window without ever having been exposed appears in no population; the complement is over the *live* corpus. That the inverse exists (the 152 served and later deleted) proves there is churn within the window. The direction of this bias is unknown and cannot be measured with the data we have.

4.1.1 Aggregate capacity does not force the outcome — and what this does not establish

Caveat: one objection is that 583,763 accumulated slots are not fungible. The surface delivers 10 items per session, and if a session needed more than 10 relevant items, capacity would be binding *today*, however much slack there was in the aggregate. The objection is correct and restricts the claim: we do not claim that 10 slots per session are many.

The claim is about rotation, not session size. The question measured is how many *distinct* items the surface has ever shown, and for that question the slots are fungible over time: nothing obliges today’s session to show the same 10 items as yesterday’s, and showing ten different items per session would never violate the limit of ten.

And the evidence that aggregate capacity does not bind is one line of arithmetic, not a test: 583,763 slots against 67,187 chunks. There was room to show everything, eight times over. That is what can be asserted, and it suffices for what the previous paragraph supports.

Note: this section previously carried a table of “opposing predictions” (ratio `slots/distinct` ≈ 1 versus $\gg 1$) presented as the test that separated the two hypotheses. It was withdrawn; the argument is in Appendix F-1.

Note: this is why the uniform counterfactual (99.98%, the expected coverage under independent uniform draws with replacement; a rotation over the same slots would reach 100%) appears in this paper as a reference point and not as a recommended policy. Serving memory at random would be worse than not serving; the number exists to say what capacity *would allow*, not what should be done.

The two surfaces are not the same kind of thing. The brief proactively selects content for delivery; the search counter records candidacy in tracked calls without identifying their initiators. Of the 10,899 live chunks with an exposure record, 9,755 have a positive search counter, an upper bound on what tracked search returned (§3.1). The counter records that a chunk was a top candidate of some tracked search call, whoever initiated the call (automated callers included) and whether or not the chunk was then returned; it does not say how much of it reached an agent. The brief (1,635 live; 1,787 counting the 152 later deleted) records what was selected for delivery. This does not invalidate the complement (“never exposed” remains absence of a record on both, a lower bound on non-delivery by the brief and by tracked search), but it restricts what can be said about cause: the 83.78% figure measures what has no record of arriving, not what the ranker refused. The claims about mechanism (§5) hold for the brief.

Caveat: the two surfaces also intersect, which the decomposition “9,755 came from search, 1,787 from the brief” hides: $1,787 + 9,755 = 11,542$ against a historical union of 11,051 $\Rightarrow$ 491 chunks are in both. The two numbers do not partition the 10,899, and presenting them side by side as if they did is the same mixing of universes that erratum F-3.2 corrected elsewhere. Added on 2026-09-21 following adversarial review.

Caveat: the two rows count different populations, and the sum gives it away: $11,051 + 56,288 = 67,339$, 152 more than the corpus. The historical union combines historical brief-log membership with positive search counters on live chunks; it includes 152 chunks served in the brief and deleted afterwards and does not count delivered exposure. The complement counts live chunks with neither record at 2026-08-28 09:52Z (`out/superficie.json`), a lower bound on non-delivery by the brief and tracked search, not across all agent-facing search. Discounting them, $10,899 + 56,288 = 67,187$ exactly. The percentage cited is over the live corpus, over which this record predicate is measured.

4.2 Secondary result: the large types cluster at 10.7–27.0% exposure, and size cannot be separated from curation

Caveat. “Exposed” here is the UNION of the two surfaces (`brief_log` $\cup$ `access_count` > 0), the same definition as in §4.1. This needs saying because most of the records are search-counter records, which mark candidacy in a tracked search call, not delivery and not who initiated the call (§3.1). The gradient below describes that search candidacy added to what the brief delivered. For live chunks the two contributions could be tabulated separately by type (brief-log membership and `access_count` > 0 are both recorded per chunk, with an intersection); that split is not reported here.

type	exposed/total	%
lesson	53/53	100.0
test	14/14	100.0
project	36/43	83.7
feedback	12/17	70.6
digest	15/25	60.0
person	8/14	57.1
decision	4/11	36.4
shared	13/40	32.5
graph_node	282/1,046	27.0
daily	798/3,231	24.7
team	3,327/15,308	21.7
distilled	2,822/14,456	19.5
other	3,515/32,920	10.7

Caveat. The table above uses a filter that is in the code and was not in the text, and it is still in the code. `superficie-de-exposicao.py` selects types with `HAVING total >= 10`, which excludes `pending` (n=6) and `procedure` (n=3). Both have 0% exposure and both are small; that is, the filter removes exactly the evidence that contradicts “small $\Rightarrow$ highly exposed”. A filter that can only help must be declared, and its effect measured (`measurement/robustez-tamanho-exposicao.py`).

Two explanations compete: curation (more curated types are more exposed) and size (small collections fit on the surface). Nothing here separates them ($\leftarrow$ F-3.5). The tests below separate *filter artifact* from *signal*; the partials further on control for age, mean importance and text length, and none of them is curation. No variable in the corpus measures curation, so “size” and “curation” remain confounded by construction. What does change with the count is the strength:

analysis	13 types (with filter)	15 types (without filter)
Pearson r ($\log_{10} n \times \% \text{ exposed}$)	−0.728	−0.334
Spearman ρ	−0.687	−0.098
binomial β (logit, weighted by n)	−0.982	−0.961

The correlation of percentages is fragile to the filter; the binomial model is not. The reason is that correlating percentages gives a type of 3 chunks the same weight as one with 32,920. The binomial model uses all the types and weights each by the information it carries: the two excluded types move the coefficient by 2%, not by half. The binomial is what this paper reports, and the correlation stays as a description of the figure, not as a test.

$\beta = \mathbf{-0.961}$. Correction. But “ $\times 0.38$ per decade of size” is a parameterization the axis does not support, and the previous version reported it that way. The 15 types are not distributed along size: they form two clouds with a void between them (`measurement/lacuna-no-eixo-de-tamanho.py`, `out/size-axis-gap-2026-08-29.json`).

types with $n < 100$	10 (from 3 to 53)
types with $100 \leq n < 1,000$	0
types with $n \geq 1,000$	5 (from 1,046 to 32,920)

largest gap on the $\log_{10}(n)$ axis
— as a fraction of the axis range

1.295 decades, between `lesson` (53) and `graph_node` (1,046)
32.1%, without a single point

A slope fitted over two clouds is, arithmetically, the difference between them divided by the distance between them. It describes the data, but it does not license a point reading inside the void: a prediction such as “a type of 300 chunks would have $\times 0.38$ the chance of one of 30” has no observation at all in this corpus. With $n = 15$ independent units and a third of the axis empty, the defensible form of the finding is the contrast between the clouds, not the rate: the 5 large types occupy 10.7–27.0% of exposure; of the 10 small ones, 7 are above 32.5%, one is exactly at 32.5%, and 2 are at zero.

And the model’s standard error is unusable. It gives ± 0.027 ($z = -35$) because it assumes 67,187 independent observations; the predictor is constant within the type, so the unit of independence is the type and the effective n is 15. By the jackknife over types: $SE = 0.471$, $z = -2.0$. Leaving one type out moves β within $[-1.12, -0.51]$. The finding survives, but narrowly. Saying the opposite would be selling as robust a result that a single collection can nearly halve.

Corollary. The same argument imposes this: recomputing the correlation “at the chunk level” does not answer the ecological objection. Since $\log_{10}(\text{size})$ does not vary within the type, the point-biserial r over 67,187 chunks (-0.150) merely reweights the same 15 points. Disaggregating a predictor that is a property of the group does not create information.

What survives intact, and is the strongest form of the finding: *no small type falls within the range of the large ones*. The 5 types with $n \geq 1,000$ occupy 10.7–27.0%, and none of the 10 types with $n < 100$ is in that interval; they are above it (32.5% to 100%) or at zero. Correction. The phrase “no overlap” in the previous version was stronger than that and false with the 15 types: including `pending` and `procedure`, the range of the small ones becomes 0–100%, which contains that of the large ones entirely. The two cases at zero are explainable (`pending` was 6 days old at the measurement instant, 2026-08-28; `procedure` has 3 chunks), and the explanation is *post hoc*; it is here as a limit, not as a defense.

And the age confounder does not explain the finding. Large types are, in fact, older ($r(\log n, \text{age}) = +0.41$), but the partial controlling for age stays at -0.709 , almost unchanged. Restricting to the 9 types with mean age ≥ 70 days, where age is approximately constant, the relationship strengthens: $r = -0.843$, $\rho = -0.883$.

These partials are over the 13 filtered types, and therefore inherit the fragility that the previous paragraph just established. “Almost unchanged” is relative to -0.728 , which is the Pearson with the filter; the Pearson for the 15 is -0.334 , and against it -0.709 would not be “almost unchanged”: it would be double in magnitude. The “9 types with age ≥ 70 days” are also 9 of the 13 ($\leftrightarrow$ F-3.6). What they establish, then, is limited: within the filtered subset, age does not explain the gradient. They are not evidence about the 15. The same holds for mean importance (partial -0.685) and for text length (-0.732).

And the confounder that remains after age is what prevents the strong conclusion. §4.3.1 shows that ingestion arrives in batches of files: large types are those fed by an automatic pipeline; small types are written by hand. This is curation disguised as size, a common cause of both, which no correlation between them can separate. The paper does not rule out the curation hypothesis; what it shows is that size predicts better than the *relevance the system itself assigns*, which is a weaker claim and is the one the data support.

Two properties of the corpus composition limit how surprising the separation is. There is no type with n between 53 and 1,046. With a void in the middle of the axis, some separation between “large” and “small” is guaranteed by the composition, not discovered. And `other` alone is 49% of the corpus, so the aggregate of 83.78% is, in large part, a single type.

`lesson` is at 100% and has 53 rows. The relevance assigned by the system predicts exposure less well than the size of the collection does, with the caveat above that size may be a proxy for how the type is produced.

Figure 1. `out/fig1-capacidade.svg`: scatter of $\log_{10}(\text{size}) \times \% \text{ exposed}$, one point for each of the 15 types, with the two bands ($n \geq 1,000 \cdot n < 100$) shaded and the stretch of the axis without observations

marked. Generated by `fig1-capacidade.py --dados out/SIZE-EXPOSURE-15-2026-08-29.json`, a locked artifact.

Correction. Three design decisions, and all three are retractions of earlier versions of the figure. (i) It was generated over the artifact with `HAVING total >= 10` and showed 13 types, and the two omitted, `pending` and `procedure`, are exactly those that sit at 0% and break the reading of a clean separation. A figure that visually asserts what the text refutes is worse than no figure, because the reader believes what they see before reading the caveat; a caveat placed below the figure does not fix this, for the same reason. The guard in `fig1-capacidade.py` now refuses the filtered artifact. (ii) The regression line was removed: it crossed 1.295 decades without a single point, which asserts graphically the continuity that §4.2 withdraws from the text. In its place, the stretch is shaded and labeled *no observation*. (iii) With the 15 types the Pearson of the figure is -0.334 , not -0.728 . The figure shows the fragile number, the one computed without the filter, and the binomial model stays in the text.

What the figure shows, then: the 5 large types cluster at 10.7–27.0%, while the 10 small ones spread across 0–100%. Rather than a separation, this is radically greater variance in the small ones, with the range of the small ones containing that of the large ones.

4.3 The brief surface is a carousel of 201 items

Closed window [2026-08-20, 2026-08-27):

slots served	46,295 in 4,632 briefs
— deficit against 10 per brief	$25 = 4,632 \times 10 - 46,295$
distinct chunks	201, of which 5 were served only by our five health probes; 196 organic
present in 100% of briefs	3
top-10	47.16% of slots · top-20 61.46%
without the 5 probes	46,270 slots in 4,627 briefs · deficit 0 · top-10 47.16% · top-20 61.46%

Figure 2 (`out/fig2-concentracao.svg`): Lorenz curve of the carousel (rank $\times$ cumulative share of slots), with marks on the 3 constant items (which alone take 30.0% of the slots) and on the top-10 cut. Generated by `fig2-concentracao.py --dados out/superficie.json`, and the script aborts if the curve does not sum to `slots_7d`.

Caveat: the 25 missing slots, and what they suggest (added 2026-09-21, following adversarial review). An earlier version of §2 said the brief comes *“always with 10 items”*, and $4,632 \times 10 = 46,320$ against the 46,295 measured. §5.7.2 establishes that the health probes are briefs of 5 lines and that there are five of them, and $5 \times 5 = 25$, exactly the deficit. The arithmetic suggests that the 4,632 include the five probes, that is, that the denominator of this section carries the same contamination that §5.7.2 turns into a headline elsewhere.

Confirmed by identity, not only by arithmetic (2026-10-04; `measurement/sprint-recon-52-e-sondas.py`, `RECON-52-e-sondas-2026-10-04.json`). The window holds exactly 5 briefs that do not have 10 rows. Each has 5 rows; the other 4,627 all have 10, and no row lacks a `brief_id`. Their `brief_ids` are, as a set, the five that `out/ancora-sondas.json` lists as `sondas_excluidas`. All are `scope=global` with no agent, issued on 2026-08-26 at 19:58:17–18 and 20:28:55–56 UTC. So the 4,632 briefs do include our five health probes, and $25 = 5 \times 5$ is an identity. The probes are also not neutral to the distinct count. Each carries the three constants plus two other chunks, and five of those chunks (307967–307970 from `memory/decisions.md`, 308443 from `memory/projects.md`) were served exactly once ever, inside a probe. Without the probes the window has 46,270 slots in 4,627 briefs and 196 distinct chunks. Top-10 and top-20 shares are unchanged (47.16%, 61.46%), and the same 3 chunks appear in 100% of briefs. The earlier phrase *“always with 10 items”* was false for exactly these five probe briefs; §2 now states the exception.

The 201 are what was measured, not the ceiling. There was room for 46,295, one per slot. Treating the measured value as the ceiling would turn a result into a pigeonhole and teach the reader to find the curve

inevitable. The diagonal is a reference for reading; the finding is that the curve is this far from it with a $230\times$ margin by which it need not have been. (F-2)

4.3.1 The carousel does not rotate because there is nowhere for it to rotate

The daily diversity of the coverage channel has steps. The explanation we tried first, that “each ingestion batch feeds the channel for seven days and expires”, was refuted by a dated prediction that we registered before the day (§6 and PREDICTION-2026-08-29.md). The refutation exposed the structure described below.

The channel sees 0.16% of the corpus. It is the union of two sub-pools, each with its own path patterns and its own window (brief.ts:135,642,645; brief-diversity.ts:59-62):

sub-pool	patterns	window	eligible
per agent	sessions/<agent>/%	7 d	0
global	memory/entities/%, memory/lessons.md	30 d	108

The zero in the per-agent sub-pool does not come from a lack of material. On 2026-08-30, 10,899 sessions/% chunks pass the importance floor, and none falls within the 7-day window, because the newest of them is 20.5 days old. Caveat: this 10,899 has no relation to the 10,899 of the live union of exposed items in §4.1: they are distinct sets whose sizes coincide at this instant. We checked: the intersection between them is 1,971, and the live union measured on the same day is 10,926. The coincidence belongs to the calendar, not to the quantity, and disappears the next day. (The 20.5 days, the 1,971 and the 10,926 have no preserved artifact.)

The system did have relevant session memory; none of it was younger than the sub-pool’s 7-day window. We later established why (DEVIATIONS-FOR-PAPER.md §10.5, §10.7). A host migration had moved both the address and the schema of the agents’ session transcripts, and session ingestion produced zero chunks from 2026-08-11 until it was repaired on 2026-09-07, while its cron reported ok. Separately, all ingestion was frozen from 2026-08-24 01:06Z to 2026-09-07 14:06Z by a watcher invoking a missing binary. Half of the coverage channel was therefore inert because of an ingestion defect, not by design and not because of the normal cadence of ingestion. After the repair, the per-agent sub-pool held 285 eligible chunks (2026-09-09). The served coverage set did not follow. From the serving log alone (_sprint-2026-10-04/A-rc2/coverage-set-from-log.py, A-rc2/COVERAGE-SET-FROM-LOG-2026-10-04.json), the ids served outside the main pool are the same 108 ids (spanning 308214 to 308496, not contiguous) on every day of the log from 2026-08-23 to 2026-09-19 (2026-09-02 has no rows), the repair of 2026-09-07 included; the set first changes on 2026-09-20. Why the post-repair 285 never reached the served set is not established here, and neither is how it relates to the pool of 55 that a copy frozen on 2026-09-08 gives (below). Every coverage-channel number in this section was measured inside that broken regime, including the 108-chunk pool and its daily exhaustion on 2026-08-26 to 2026-08-29; so were the five days without new items that §1 cites. They describe the channel under defective ingestion, not its steady state.

Caveat. This makes the composition of the channel unstable in a way that the table above hides: a burst of new sessions would repopulate the per-agent sub-pool and change the eligible pool (and with it the tie structure that sets the ceiling) without a single line of code changing. The obligation to say this had been registered in DEVIATIONS-FOR-PAPER.md since 2026-08-27 and had not been fulfilled; the 2026-08-30 retrospective found it by cross-checking the registered commitments against what the manuscript carries.

Both are subject to the importance floor $\text{importance} \geq 0.7$ OR $\text{pain} \geq 0.7$. Measured over four closed days (measurement/pool-elegivel.py, POOL-ELEGIVEL-2026-08-28.json; the four days are in POOL-ELEGIVEL-2026-08-26-to-29.json, produced by measurement/sprint-pool-elegivel-multidia.py):

Caveat: until 2026-10-04 the artifact covered ONE day while the claim covered four. Adversarial review on 2026-09-21 found that POOL-ELEGIVEL-2026-08-28.json holds only 2026-08-28 ("dia": "2026-08-28", "dia_parcial": false). The other three days had been measured by the same route and not preserved, which is exactly the gap that the backing-artifact discipline of §6.1 exists to prevent: an artifact cited to support more than it contains. The review suspected in-place rewriting; there was none, because the file on disk is identical to the versioned one. The gap was closed by re-measuring from a preserved trial database

(corpus-SERVING-REAL-e20260903-recuperado.db, sha256-verified copy) using the same predicate. A positive control came first: re-running 2026-08-28 reproduces all 12 fields of the original artifact. 2026-08-26, 2026-08-27 and 2026-08-29 then give the same pool of 108, 0 never served and 108 served (POOL-ELEGIVEL-2026-08-26-to-29.json). The set of pool ids is identical (same hash) in copies frozen on 2026-08-26, 2026-09-03 and 2026-09-07. A copy frozen on 2026-09-08, after the corpus was re-ingested, fails the control (pool 55) and is not used. Limit: importance and pain are read as stored in those copies, so they bracket the four days rather than observe each instant.

live corpus	67,187
eligible pool of the channel	108 — 0.161%
of those, never served	0
served on the day	108 — 100% of the pool, on 2026-08-26, 2026-08-27, 2026-08-28 and 2026-08-29
coverage slots per eligible candidate	12.4× (closed day; 12.5× on 2026-08-26, which had 5 extra five-row briefs)

Caveat. The last number is for a closed day. The original 2026-08-29 reading was taken with the day still in progress and gave 5.6× only because fewer briefs had occurred; that partial reading was not preserved. Measured on the closed day, 2026-08-29 gives 12.4× (672 briefs). The script flags `dia_parcial` so that partial and closed days are not compared.

The pool is exhausted on every day measured. There are twelve times more slots than candidates, so the ordering `last_served` ASC orders but does not exclude anyone: every eligible item appears, every day. There is no carousel turning and no batch expiring; the pool is too small for rotation to be a question.

This changes where the scarcity lives. On the day, coverage of the eligible pool is 100%. In one brief, there are 2 slots for 108 candidates. What the ordering decides, and what the intervention of \$5 moves, is *which two appear in each brief*, not *how much of the corpus is reached*. The corpus is not reached because 99.84% of it never enters the draw.

Why the prediction failed. There are two coverage sub-pools, with different patterns and windows:

sub-pool	patterns	window
per agent	<code>sessions/<agent>/%</code>	<code>freshMaxAgeDays = 7</code>
global	<code>memory/entities/%</code> , <code>memory/lessons.md</code>	<code>freshGlobalMaxAgeDays = 30</code>

The batch of 2026-08-09 to 2026-08-10, on which we made the retrodiction, was entirely `sessions/boris/...`, the per-agent sub-pool, so it stopped cleanly at 7.0 days, with the maximum age served never reaching 7.00 in eight days. The batch of 2026-08-21 to 2026-08-22, on which we made the prediction, is `entities/% + lessons.md`, the global sub-pool, with a 30-day window. We applied the window of one channel to a batch of the other. The retrodiction was valid; the extrapolation was not. It is the same defect of “verifying an invariant over the wrong set” that §6 catalogs, committed this time on the very mechanism that the paper describes.

Attribution by channel was measured, not inferred. The decisive objection to everything above is that `brief_log` does not record the origin of each serve, and an eligibility rule excludes but does not attribute: the same 108 serves would be equally compatible with “coverage stopped at 7 days and the main pool served everything”. The test that separates the two hypotheses needs no new column. It suffices to run the same state twice through the real code, once with `freshSlots = 2` (production) and once with `freshSlots = 0`, and take the difference. What disappears when the channel is switched off is, by construction, what the channel delivered.

In 40 briefs of 2026-08-29 (`CHANNEL-ATTRIBUTION-2026-08-29.json`): 80 slots attributed to coverage (exactly 2 per brief, across all 40), 62 distinct chunks, and all 62 belong to the 2026-08-21 to 2026-08-22 batch. Coverage *was* serving the batch at 7.4 days; the alternative hypothesis is ruled out by measurement.

Caveat: without this test, the argument was asymmetric. `brief_log` does not record through which channel each line was served; there is no origin column. Counting serves of a batch measures the union of the main

pool with the coverage pool, and the main pool has no age filter at all. That is why the cut guard flagged “age 7.42 served” as a violation of the 7-day window: part of those serves was never subject to any window. Using this limitation to void the old guard without applying it to the new explanation would be choosing the ruler by the result, and the new explanation suffered from the same union bias until the differential test above existed.

This is a result in its own right, and it concerns the channel’s daily reach, not its per-brief selection. Eligibility combines path patterns, the importance/pain floor and age windows. In the measured regime, the channel exhausted that eligible population on every measured day, and the corpus grew outside the configured path patterns. Holding eligibility fixed, an additive salience bonus changes per-brief selection only within `last_served` ties; the tested intervention saturated at 17/350 replay states (§5.3, §5.4).

The main pool, filter by filter. The main pool reaches its 8 slots through seven serial stages (`brief.ts:202-215, 371-397, 420-468, 819-821`): (F1) routing, where the agent sub-pool sees only `sessions/<agent>/%` with quota 5 and the `scope=global` sub-pool sees the whole corpus; (F2) a `since` window, inactive because production sends none; (F3) a SQL proxy pre-rank with `LIMIT 500` per sub-pool; (F4) the exact `calculateSalience` re-rank; (F5) dedup by id, exact title/one-liner and near-duplicate containment ≥ 0.6 ; (F6) quotas capped at `mainTarget = n - freshSlots = 8`; and (F7) the high-pain floor, which picks the items of the no-fresh brief with `pain  $\geq 0.9$` . F7 is listed last but runs first: it is phase 0 of `pickDedup`, and the pinned items it places count toward `mainTarget` before the F6 quota pass (phase 1) and the backfill by global score (`brief.ts:438-480`). We reimplemented them on the frozen corpus of these days (`p2-ord-ro-2026-08-26.db`, 67,187 chunks; `measurement/sprint-desagrega-filtros-pool-principal.py`, `A-filters-disaggregation/out-ord0826.json`). The reconstruction reproduces the served main slots in 100% of the briefs of each day with rows from 2026-08-24 to 2026-09-07 (672 per day; 441 on 2026-09-03; 2026-09-02 has none), counting a brief as reproduced when its 8 reconstructed ids lie inside the 10 served and leave the 2 coverage slots. Counting exclusions is the wrong instrument here: F3 alone removes 95.7% of the corpus from candidacy and binds nothing. The question is answered by lifting one filter at a time and recomputing every brief:

lifted	briefs whose main set changes (of 672)	distinct main chunks per day
none (as served)	—	33
F1 routing	672	8
F2 <code>since</code>	— (inactive)	33
F3 <code>LIMIT 500</code>	0	33
F4 exact re-rank	672	33
F5 dedup	192	33
F6 quota split	672	8
F7 high-pain floor	672	33

No filter, lifted, widens the surface; the two that change its breadth narrow it, because routing by agent is the main pool’s only source of variety. The breadth is exactly $3 + 6 \text{ agents} \times 5$: the 3 shared slots are taken by the high-pain floor (chunks 112241, 116107 and 116467, `pain = 1.0`, in every brief), and each agent gets the same 5 in every brief. A day has 5,376 main slots and they serve 33 distinct chunks, which is 0.61% of the slot capacity and 163 serves per chunk. From the serving log alone, without the database (`observed-main-from-log.json`), there are 33 distinct main chunks on every measurable day from 2026-08-21 to 2026-09-21 except 2026-08-23, which shows 34, and the set is identical on every measurable day from 2026-08-24 to 2026-09-19. On 2026-08-21 and 2026-08-22 one `boris` slot carries 285042 instead of 298048, 2026-08-23 carries both, and on 2026-09-21 three ids (309422, 309529, 311215) replace three (273772, 285044, 298048). The month’s union is 37 (33 + 285042 + the three new ids). A rotation over the same slots could serve 5,376 distinct chunks a day, or 8.0% of the corpus. So the main pool is not capacity-bound at the day, the unit at which §4.1 measures exposure. It is bound by its ranking: a deterministic top-k with no serve-history term, removed on purpose on 2026-06-26 (`brief.ts:786-792`). Both channels are therefore governed by policy rather than by slots, but by different policies. The coverage channel exhausts an eligibility predicate of 108 chunks; the main pool returns the same arg-max for the same route, with no eligibility predicate: all 67,187 chunks enter the SQL pre-rank, and 99.95% of them are never reached. Daily, the two channels are disjoint and add up to

the whole: $33 + 108 = 141$ distinct chunks, every measurable day from 2026-08-24 to 2026-09-19 (2026-09-02 has no rows).

Caveat: a counting trap, recorded because it nearly caught us. Counting exposure with JOIN **chunks** makes 2026-08-21 show 33 distinct chunks where **brief_log** says 85. The other 52 no longer exist. The same 52 vanish from 2026-08-20 ($87 \rightarrow 35$) and 2026-08-22 ($193 \rightarrow 141$). They are one ingestion of **memory/lessons.md** (ids 308114–308165), replaced by a re-ingest on 2026-08-22 02:01:58 UTC, and the serving path kept handing them out for about 17 h afterwards (966 of their 2,600 slots; **RECON-52-e-sondas-2026-10-04.json**). This trap was first recorded here as 2026-08-20 with $85 \rightarrow 33$; measured per day (UTC and BRT alike), that pair belongs to 2026-08-21. Exposure counts come from **brief_log**; only what needs chunk metadata uses the JOIN, and it declares the loss.

4.3.2 The other channel also freezes, for the opposite reason

If the coverage channel fails because of the calendar and because of algebra, the thesis still needs to explain why the other 8 slots concentrate. There the score is the dominant coordinate and an additive bonus has no ceiling, so the concentration cannot be explained by a bounded response to the score. It has another cause, and it is read directly off the formula.

$$\text{salience} = 0.55 \cdot \text{importance} + 0.15 \cdot \text{recency} + 0.10 \cdot \text{pain} + 0.20 \cdot \text{access}$$

Of the four terms, three do not decay. Importance and pain are static. The access term is $0.20 \cdot \log_{10}(\text{access_count}) / \log(1000)$ over a monotonic counter: it rises and never falls. Recency is $2^{-(\text{age}/\text{retention})}$, with age measured from **last_accessed_at** when it exists and from **source_date** otherwise, and it has no floor (**serving-salience.ts:78-98**). A NULL **retention_days** is the code’s never-decay marker (recency = 1.0, :58-71); 39,130 of the 67,187 live chunks carry it, among them 35 of the 149 served in the window and all three chunks below. For those, no term decays: the score of an old and once-popular chunk is non-decreasing and stays at its ceiling. For chunks with a positive retention, recency does decay, and a search hit resets it through **last_accessed_at**, so access enters the score twice (**out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json**, **retention_days_null**).

The three chunks present in 100% of the 4,632 briefs of the week are exactly that:

chunk	type	importance	pain	access_count	last access
112241	team	0.80	1.00	414	2026-05-30
116107	team	0.80	1.00	363	2026-07-29
116467	team	0.80	1.00	911	2026-07-17

(The dates are read from a copy of the 2026-09-07 trial epoch, **out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json**, **the_three**. **last_accessed_at** only moves forward, so a value earlier than 2026-08-28 in that copy is also its value on 2026-08-28.)

They had last been accessed 90, 30 and 42 days before 2026-08-28, and they won every brief of that week. The positions they occupy are determined by tracked search traffic from months ago, whoever initiated it (§3.1), and “determined” here is a measured counterfactual, not a reading of the formula.

Caveat: the scope of the word, corrected on 2026-08-30. A read-through observed that “the top of the brief is determined” generalizes from the three chunks that the counterfactual moves to the entire top, and that the support is a recompute over 149 of the 201 chunks of the window. What is measured is the position of those three: zeroing the access component takes them from 1/3/4 to 44–46 among 149 served (23–47 across instants and access-timing assumptions). It is a claim about them, not about the ten positions.

chunk	importance	pain	accesses	position	position with access zeroed
116467	0.80	1.00	911	1	44–46 (tie at 0.69)
112241	0.80	1.00	414	3	44–46 (tie at 0.69)
116107	0.80	1.00	363	4	44–46 (tie at 0.69)

Recomputing the salience of the 149 chunks served in the window with the access component at zero, through the production function `calculateSalience` imported from `serving-salience.ts` (`measurement/sprint-contrafactual-salience-producao.mjs`, `out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json`), the three leave the top-10: they fall from positions 1, 3 and 4 to a three-way tie at 0.69, at positions 44–46 of 149 on 2026-08-29 (45–47 at noon of each day of the window). The epoch keeps only the latest `last_accessed_at`, so competitors accessed after the pinned instant get full recency; dropping those accesses gives 23–25. Across the tested instants and access-timing assumptions, positions span 23–47 in this fixed retrospective population, with every other attribute as in the copy: a sensitivity range, not a reconstructed historical serving position. On the path the brief uses (the `scope=global` pool: SQL pre-rank, LIMIT 500, `calculateSalience` re-rank, `brief.ts:359–398`) they fall from pool ranks 1, 5 and 6 into a tie at 0.69. With the access count zeroed (`prod_acc0`), 76–79 candidates from 43–46 distinct source files score strictly above them; with no access history at all (`prod_noacc`, which also drops `last_accessed_at` from recency), 57–79 candidates from 24–46 files do. For agent-specific requests that pool has a nominal quota of five in the quota pass (phase 1 of `pickDedup`), which runs after the pinned high-pain items have been placed (phase 0, counting toward `mainTarget`) and before the backfill by global score; without an agent its quota is the whole brief (`buildPools`, `pickDedup` in `brief.ts:438–480`). Their positions (95–119 under `prod_acc0`, 76–119 under `prod_noacc`) are stable-sort order inside the 0.69 tie block, not a ranking. The recomputation stops before final selection: near-duplicate dedup was not replayed, and it may remove candidates above them or reject them, so these pool positions do not by themselves establish final served membership. The first version of this counterfactual (`contrafactual-do-topo.py`, `out/TOP-COUNTERFACTUAL-2026-08-29.json`) used a linear 365-day recency from `source_date` and a 0.5 default for `importance`, which is not the production function; it gave 131/129/128, and the new artifact reproduces that number as an anchor.

Caveat: the population of the counterfactual is 149, while §4.3 counts 201 distinct chunks in the same window. The 52 missing are one identified set, not a coincidence (an open reconciliation declared on 2026-09-21 after adversarial review, closed on 2026-10-04 by `measurement/sprint-recon-52-e-sondas.py`, `RECON-52-e-sondas-2026-10-04.json`). As sets, the 52 absent from the week (201 → 149) are exactly the 52 absent from each of 2026-08-20, 2026-08-21 and 2026-08-22. They are ids 308114–308165: one ingestion of `memory/lessons.md` (`lesson`, created 2026-08-20 02:02:03 UTC), served only by the coverage channel (2 per brief, 1,300 briefs, all six agents) from 2026-08-20 21:38 to 2026-08-22 19:07 UTC. The file was re-ingested on 2026-08-22 02:01:58 UTC and the 52 were replaced by 53 new chunks. All 53 replacements were served in the window and are among the 149. A deleted chunk has no `importance`, `pain` or `access_count` to recompute salience from, so the 149-chunk recompute of 2026-08-29 leaves them out. The 52 were restored from a backup taken while they existed; the result is below.

Caveat: the set is also defined by the original score, which the adversarial objection pointed out as possible circularity (correctly, in form). The choice is conservative, by an argument that does not depend on measurement: recomputing over a larger pool can only add competitors, and adding competitors never improves anyone’s position. So 44–46 is a lower bound on the drop. It is now measured on the full pool (`RECON-52-e-sondas-2026-10-04.json` for the restored rows, `out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json` for the scores): with the 52 deleted chunks restored from a backup taken while they existed (2026-08-21 06:00 UTC), and zeroing the access term, the three fall to 54–56 of 201. That is 10 places lower, not 52: of the restored chunks only the 10 with `pain` = 0.9 outrank them. One caveat: the 52 are earlier versions of `lessons.md` whose replacements are already among the 149, so the 201-pool counts that content twice; it is nonetheless the population that was served. Excluding our five health probes (§4.3) the figures are 40–42 of 144 and 50–52 of 196. Every variant takes the three out of the top-10; none places them beyond rank 100.

Caveat. The objection that motivated this measurement, raised in adversarial review, was that `pain` = 1.00 and `importance` = 0.80 could suffice to put them at the top, making access irrelevant and the word “determined” too strong. The objection was valid in form and false in substance: those values are high, but not unique. Between 22 and 44 other chunks of the 149 score strictly above them without the access term, and it is the old traffic that puts the three ahead of them.

Caveat: a record of how the claim came to need this. The sentence said “the top of the brief is a *fossil* of search traffic from months ago”, and the metaphor was replaced by “determined by” in a pass that removed value judgment from the text. But “determined” is a stronger causal claim than the metaphor it replaced:

neutralizing the tone hardened the claim. The counterfactual above exists because the swap left it exposed. Nor is it an isolated case: of the 9,755 chunks with any access, 7,908 (81%) had gone more than 60 days without being accessed, with the access component intact. This count has no preserved artifact, and its instant is not recorded; by context it is the 2026-08-28 measurement.

Note: the three constant items belong to the main pool, by deduction. The coverage channel orders by `last_served` ASC. A chunk served in the previous brief has the most recent possible `last_served`, so it sits at the end of that ordering, behind the entire stratum of never-served items. A chunk present in 4,632 of 4,632 briefs cannot have been chosen by a comparator that prioritizes the least-recently- served. This decomposes the concentration between the two channels without needing a position column in the log, which does not exist.

Caveat: the feedback loop is NOT closed by the system, and this is a deliberate design decision. `access_count` is incremented only in `search.ts:396`, and the brief declares in its header that it is “*read-only over chunks; does NOT touch access_count*”. So serving in the brief does not raise the priority of anything, which separates this case from the classic recommendation feedback loop (Chaney et al.), in which exposure reinforces itself. Here exposure in the brief does not reinforce itself; what exists is a permanent encoding, without decay, of past traffic. If the loop closes, it closes through the agent (which sees the item and may search for it again), and that we did not measure.

The symmetry that completes the thesis. The two channels of the surface freeze, for opposite reasons and neither of them tied to capacity:

channel	slots	why it freezes	responds to score adjustment?
main pool	8	deterministic score with a monotonic, non-decaying component	yes — and nobody adjusts it
coverage	2	eligible population of 108 chunks (0.16% of the corpus), exhausted 100% on each measured day (defective-ingestion regime, §4.3.1); lexicographic order	only up to a ceiling of 4.86% (§5; 2026-08-26 corpus, defective-ingestion regime)

The main pool *could* be corrected by score, and nobody corrects it. The coverage channel, designed to compensate for the main pool, has a daily reach bounded by its eligible population (path patterns, the importance/pain floor and age windows), which it exhausted in the measured regime, and within a brief it responds to score only within `last_served` ties (17/350 here) and then saturates. That is why the 8.7× headroom does not turn into coverage.

4.4 The deductive prediction, and the test

The coverage comparator is lexicographic: when `last_served` differs, `salience` is never consulted. An additive bonus on `salience` only decides within strata of identical `last_served`. Hence any intervention of this kind has a ceiling. Under the premises of §5.3 (prefix selection, filtering that does not depend on the bonus), only briefs in which the cut of the coverage pool falls inside a `last_served` tie with a designated item on the non-selected side can change, so their fraction bounds the alterable briefs from above; it is a necessary condition, not a sufficient one (§5.4). The ceiling reported here is measured directly, as the fraction of states whose served set changes at a saturating dose in a replay of the production pipeline, deduplication included (§5.6).

The test is a counterfactual replay with increasing dose over 350 real brief states, faithful to the serving pipeline. The `w = 2` column is the dose configured for the *shadow* arm during the measured window: the treated composition under it, over these 350 historical states, was computed and logged and never delivered to any agent. (In the later trial, `w = 2` was served in six epochs, as recorded in `DEVIATIONS-FOR-PAPER.md` §10.29. That is Paper B’s data, not these 350 states.) Caveat. Until 2026-08-30 that column was labelled

only (**served**), with the caveat on this line. A read-through showed that the label alone suggests delivery to the agent, because a table label is read without the paragraph that precedes it. See Appendix F.

w	0	0.5	1	2 (<i>shadow</i> dose)	4	7.5	15	100	100,000
states that change	0	5	8	11	15	17	17	17	17
displacement 0 events		5	8	12	18	20	20	20	20

- **negative control passes:** $w = 0$ gives 0 in 350/350;
- monotone in each of the 350 states, not only in the aggregate;
- ceiling $17/350 = 4.86\%$;
- saturation at $w \in (4.0, 4.4]$: on a fine grid of 23 doses (0.02 to 13) over the 17 states that move, the lowest tested dose at which a state changes ranges from 0.02 to 4.4 (median 1.7), with 0 non-monotone states and 0 states that never change on the grid. Five states already change at 0.02, the lowest dose tested, so their thresholds are bounded only from above; the spread across states is at least $200 \times (4.0/0.02)$ and is not measured exactly;
- the *shadow* dose of the measured window reproduces the published rate: $w = 2$ gives $11/350 = 3.1429\%$.

Figure 3 (out/fig3-dose-resposta.svg): the two dose-response series on a logarithmic w axis, a coarse grid (9 doses $\times$ 350 states) and a fine grid (23 doses $\times$ the 17 that move), with the registered band $\{2 \cdot 4 \cdot 7.5\}$ marked on the axis, the 17/350 ceiling line and the saturation region. Generated by fig3-dose-resposta.py.

Caveat. The two series have different denominators and the same numerator, and that is what authorizes overlaying them: a state outside the 17 does not move at any dose. This can be checked rather than argued. The series share exactly one dose ($w = 1$), and there both give 8. The script aborts if they cease to match, because two curves on the same axis that disagree where they cross make a figure that is silently wrong.

Caveat. The negative control $w = 0 \Rightarrow 0/350$ appears as an annotation, not as a point, because zero does not exist on a log axis. Placing it on the first tick would mean plotting $w = 0.015$, which was not measured; omitting it would mean not reporting the control.

Caveat. The quantity that governs is distance, not step, but this measurement does not show it. At the largest $w_{\min}$ (4.4) the two S1 items that enter receive $4.4 \cdot 0.043 \cdot 0.25 = 0.0473$, $0.90 \times$ the largest gap between adjacent candidates of the pool (0.05272, measured in a single state; out/BONUS-VS-STEP-2026-10-05.json). An earlier version said 0.0946 and $1.79 \times$, using a multiplier of 0.5 for S1 that production does not apply. Whether these items crossed one position or several is not established.

4.5 What these measurements do not identify

This section is mandatory, and it comes before the discussion of purpose. The most important limit is the first, and it holds for everything this paper reports:

- nothing here says that the missing exposure matters. There is no instrumented downstream outcome. Three agent-facing quality tables have 0 rows; this was measured on 2026-08-30 and not re-verified after the trial closed (dose switched off 2026-09-21 09:43:05Z, DEVIATIONS-FOR-PAPER.md §10.29), so it is a declared debt and not a current fact. The search telemetry recorded mostly the cron health probe: in a closed 7-day window analysed on 2026-08-27, 325 of 343 rows (94.8%) fall within one minute of the two cron firings per hour (minutes 21–23 and 51–53 of the 22,52 * * * * schedule), leaving 18 rows (2.6/day) outside those windows, whose origin was not independently established. Of the 25 columns of that table, 16 had no writer in a census taken on 2026-08-27, among them the only one with per-chunk identification (top_chunk_ids). Its writer was restored that night (commit 32f78109, together with top_scores; query_text deliberately left out), less than a day before the exposure window closes (2026-08-28, out/superficie.json), so that window is almost entirely without it. That 83.78% of the corpus has no record of reaching the agent through the brief or tracked search (a lower bound on

non-delivery; non-delivery across all agent-facing search is not established, §3.1) is a fact about the surface, not about the usefulness of what was left out. A reader who concludes “the system is losing valuable information” has gone beyond what was measured;

- nothing is randomized, and nothing needs to be. This paper does not estimate an intervention effect. Every comparison is descriptive or deductive: the surface is a census, and the ceiling of §5 is a consequence of the comparator verified by replay. Effect estimation is the object of the interventional study, which is not reported here (see “Relation to the pre-registration”);
- one system, one corpus, one operator. The exposure numbers are this system’s. What generalizes is the method and the shape of the argument (a lexicographic comparator imposes a ceiling, and the ceiling depends on the granularity of the key), not the percentages;
- **the coverage state feeds back.** `last_served` is not frozen: serving at T alters the strata structure at T+1. This limits any extrapolation of the ceiling to sustained-treatment regimes.

5. Mechanism

This section is deductive. What it proves holds for any ranker with the same form, serving a prefix of one lexicographic order, and it is the reason a study on an $n = 1$ system still says something general (§7). The whole structure comes from seven lines of code, quoted rather than paraphrased.

5.1 The object

Let P be the set of eligible candidates (the *pool*). Each $c \in P$ has:

- $\ell(c) \in \mathbb{R} \cup \{-\infty\}$: the instant of the last serve, with $-\infty$ for never-served;
- $s(c) \in \mathbb{R}$: the **salience**.

The production comparator, verbatim (`src/api/brief-diversity.ts:130-140`):

```
const a1 = aLastServedMs ?? Number.NEGATIVE_INFINITY;
const b1 = bLastServedMs ?? Number.NEGATIVE_INFINITY;
if (a1 !== b1) return a1 - b1;           // ASC: menos-recentemente-servido primeiro
return bSalience - aSalience;         // tie: maior salience
```

This is exactly the lexicographic order $\prec$ on the pair $(\ell(c), -s(c))$:

$$c \prec c' \iff \ell(c) < \ell(c'), \text{ or } \ell(c) = \ell(c') \text{ and } s(c) > s(c').$$

The intervention is a function $b : P \rightarrow \mathbb{R}_+$, zero outside the designated set D . It enters at a single point (`src/api/brief.ts:612-614`):

```
const eff = (c) => c.salience + (boosts?.get(c.row.id) ?? 0);
ranked.sort((a, b) => coverageCompare(a.lastServedMs, eff(a), b.lastServedMs, eff(b)));
```

That is: b shifts only the second argument of `coverageCompare`. Write $\prec_b$ for the resulting order.

Where D comes from, and what of that matters here. The designated set has 19 designated items, one per signature group, drawn from a population of 55 by a rule with a declared seed and verifiable precedence over the beacon (Appendix B). For what this paper measures, D is merely a fixed set that a third party can re-derive: an independent reviewer reproduced the 19 using only the public beacon and the deposited CSV. The criterion that selected the 55 is severity adjudication by a panel, and it belongs to the interventional study (Appendix C).

Caveat. One of the two numbers in §5 inherits that provenance and the other does not. The effective bonus is $W = w \cdot \Delta_{\text{cut}} \cdot \text{severity_pain}$, so the panel’s label scales the dose per chunk:

- the ceiling (§5.3, §5.7) is measured at $w = 100,000$, a dose at which the multiplier saturates along with everything else. It is independent of the severity label;
- the saturation band $(4.0, 4.4]$ (§5.4) is not: it locates a dose, and the effective dose of each chunk depends on the multiplier. Read as a property of the comparator, it would be more general than it is.

5.2 Proposition 1: invariance across strata

For any c, c' with $\ell(c) \neq \ell(c')$, the relative order of c and c' is the same under $\prec$ and under $\prec_b$, for every b .

Proof. If $\ell(c) \neq \ell(c')$, the comparator returns on the first line, whose value $a_1 - b_1$ depends on neither of the `salience` arguments. Hence b cannot alter it. ■

Define $c \sim c' \iff \ell(c) = \ell(c')$. The equivalence classes are the strata. All never-served items fall into a single stratum ($\ell = -\infty$).

Corollary 1 (the order is a concatenation). The sequence sorted under $\prec_b$ is the concatenation, over the strata in increasing order of ℓ , of each stratum sorted internally by $-s_b$. Consequently, b permutes within strata and never moves an item from one stratum to another.

5.3 Corollary 2: the ceiling, and where it lives

The surface has fixed capacity: of the 10 slots in the brief, the coverage pool feeds at most $K = \text{freshSlots}$. In production $K = 2$, and the provenance of that number has to be stated carefully, because it is not measured in the log:

- it is the default `DIVERSITY_DEFAULTS.freshSlots = 2`, overridable by `NOX_BRIEF_DIV_FRESH_SLOTS`, and there was no override, neither in the systemd unit nor in the `.env`, verified on 2026-08-27 (`SUPERFICIE-2026-08-27.md`), i.e. within the measured window. The unit's configuration changed afterwards for the trial, and it is not re-verified here;
- it is a ceiling, not a quota: the fill loop exits on `if (freshGot >= freshSlots || picked.length >= n) break (brief.ts:472)`, and a candidate rejected by deduplication does not count, so a brief may have fewer;
- and `brief_log` has no column that marks the origin of the slot, so the split “8 main + 2 coverage” is not observable in the record. It is configuration plus code, and is declared as such.

The derivation assumes that the coverage pool serves the first $\leq K$ items of $\prec_b$ that survive deduplication. In the measured window this holds: the per-agent sub-pool was empty (§4.3.1), so `interleaveFresh` returns the global sub-pool unchanged, and `pickDedup` scans it in order (`brief.ts:462-466`). It does not hold in general. With both sub-pools populated, the coverage pool is the round-robin of two orders, each sorted by $\prec_b$ (`brief-diversity.ts:180-196`); the statements below then apply to each sub-pool separately, and they do not cover the interleaving. Deduplication against the main-pool items does not depend on b . Deduplication between two coverage candidates does: when two near-duplicates share a stratum, the bonus decides which of them is served even if the cut does not fall in that stratum, and consequence 1 below does not count that case. The 17/350 is measured by replay of the production code and needs none of these premises.

The served set changes under b only if there exists a stratum that contains both a selected item and a non-selected item.

Proof. By Corollary 1, b only permutes within strata. A stratum entirely contained in the prefix has its items permuted among positions that are all selected $\Rightarrow$ the set does not change. A stratum entirely outside the prefix, likewise. What remains is the stratum that crosses the cut, and by Corollary 1 there is at most one. ■

Three consequences, and all three are measurable rather than arguable:

1. under these premises, the fraction of states in which the cut falls strictly inside a stratum and a designated item is on the non-selected side bounds the alterable decisions from above. The condition is necessary, not sufficient: with several bonused items in the stratum a designated item may not enter (§5.4). The ceiling itself is measured, as the fraction of states whose served set changes at a saturating dose in the replay, which includes deduplication. Measured in §4.4: $17/350 = 4.86\%$. In §5.6 those same 17 states are exactly the ones in which there is a swap, with all 20 entries occurring within the stratum of the item that left;

2. if ℓ were injective, $\mathbf{b}$ would have no effect at all. Every stratum would be a singleton, nothing would cross the cut, and the intervention would be identically inert. The entire room for maneuver of the intervention comes from ties in the dominant coordinate. In nox-mem these are the stratum of the never-served items and the one-second resolution of `served_at`. In the 350 measured states the first is empty (0 never-served in the pool on 2026-08-26/27, §4.3.1), so the measured 17/350 comes entirely from one-second ties;
3. the ceiling is not a design parameter: it is a property of the distribution of ℓ in the pool, which the traffic itself produces.

5.4 Corollary 3: saturation, and why step is not the right quantity

Within the stratum that crosses the cut, the order is by $-\mathbf{s}_\mathbf{b}$, and the bonus is applied to every designated item at once, so the selected item $\mathbf{c}_K$ of lowest position in that stratum may itself carry a bonus. A non-selected designated item $\mathbf{d}$ enters the served set exactly when its treated score $\mathbf{s}(\mathbf{d}) + \mathbf{b}(\mathbf{d})$, tie-break included, ranks within the slots that the stratum receives. When $\mathbf{d}$ is the only bonused item of the stratum this reduces to

$$\mathbf{b}(\mathbf{d}) > \mathbf{s}(\mathbf{c}_K) - \mathbf{s}(\mathbf{d})$$

or to equality with the tie-break placing $\mathbf{d}$ ahead of $\mathbf{c}_K$ (on equal treated scores the comparator returns 0 and the stable sort keeps the input order), so the strict inequality is sufficient but, with the tie-break included, not necessary. With several bonused items in the stratum the condition does not reduce to this form: with $K = 1$, a selected designated $\mathbf{c}$ with $\mathbf{s} = 0.8$, $\mathbf{b} = 0.4$ keeps its slot against $\mathbf{d}$ with $\mathbf{s} = 0.7$, $\mathbf{b} = 0.2$, although $0.2 > 0.8 - 0.7$. Either way the quantity is a distance to the cut, not the step to the immediate neighbour of $\mathbf{d}$. Since every bonus is proportional to $\mathbf{w}$ ($\mathbf{W} = \mathbf{w} \cdot \Delta_{\text{cut}} \cdot \text{severity_pain}$) and there are finitely many states, there is a finite dose beyond which the treated order inside every crossing stratum no longer changes: saturation is an identity, not a chosen threshold. Measured on the dose grids: the last change occurs at $\mathbf{w} \in (4.0, 4.4]$, with the lowest changing dose ranging across states from 0.02 (the grid’s lowest dose, so a censored value) to 4.4, a spread of at least $200\times$.

Caveat. This is where a monitoring trigger goes wrong. Item 7 of the registration watched $\max_j (\mathbf{s}(\mathbf{c}_j) - \mathbf{s}(\mathbf{c}_{\{j+1\}}))$, the largest step between adjacent items anywhere in the pool. That is a different quantity from the distance of §5.4, and the two are not ordered in general: the distance of $\mathbf{d}$ to the cut is the sum of the adjacent gaps along its path inside the stratum, so it is at least the largest of those gaps, while the largest gap elsewhere in the pool can be larger or smaller. As measured, the two items whose threshold is the largest $\mathbf{w}_{\min}$ (4.4; both S1) receive $4.4 \cdot 0.043 \cdot 0.25 = 0.0473$ at that dose, $0.90\times$ the largest adjacent gap of the global sub-pool (0.05272, one state at 2026-08-26 20:35Z with the inclusive cut, `out/gaps.json`), so this comparison does not show that they crossed several positions (`measurement/sprint-bonus-vs-passo.py`, `out/BONUS-VS-STEP-2026-10-05.json`). An earlier version said 0.0946 and $1.79\times$, from a severity multiplier of 0.5 for S1 hardcoded in the replay’s summary (`replay-oportunidade.mjs:944, 1014-1019`); production multiplies S1 by 0.25. A trigger on the step can stay green while the channel saturates because it watches the wrong quantity, which is what forced the reimplementaion of item 7 as the identity `churn(w_servido) = churn(w_absurdo)`.

5.5 The design consequence

A fixed-capacity surface ordered lexicographically responds to interventions on the subordinate coordinate only within ties of the dominant one, and that response is bounded by construction. To move what the agent sees beyond that bound there are three levers, and the score is not one of them:

lever	effect
the dominant coordinate (ℓ : rotation policy)	reorders across strata — no ceiling
the capacity K	moves the cut along the order: it changes which stratum (at most one, §5.3) straddles the cut and how many lie wholly inside the served prefix
the eligibility (who enters P)	changes the object, not the order
the subordinate score	bounded by Corollary 2, saturating at a finite dose (§5.4)

lever	effect
-------	--------

The table holds for the coverage channel, which is 2 of the 10 slots, and not for the whole surface. The other 8 come from the main pool, ordered by pure **salience** (§2), where an additive bonus acts on the dominant coordinate and has no ceiling. Writing “the score is not a lever” without that qualification would be false for 80% of the slots, and it is a qualification that the Abstract needs to carry, not only this section.

The correct conclusion is narrower and still holds: the channel that exists precisely to serve the never-served is the only one whose response to relevance adjustments is bounded by construction. The surface has two parts with different algebras, and the part reserved for coverage is the one that responds to the score only within **last_served** ties and then saturates (§5.3, §5.4).

It is the same conclusion as §4, reached by another route, and it has two parts that must be kept apart. The first is the channel’s daily reach, bounded by its eligible population. Eligibility combines path patterns, the importance/pain floor and age windows. In the measured regime, the channel exhausted that eligible population on every measured day (§4.3.1). The second is per-brief selection. Holding eligibility fixed, an additive salience bonus changes per-brief selection only within **last_served** ties; the tested intervention saturated at 17/350 replay states. §4 measures the first on the population of chunks; §5 proves why the second is bounded, so that, with eligibility held fixed, no additive salience adjustment could change selection there beyond the ceiling.

5.6 The test this derivation has to pass

The derivation is falsifiable. The first attempt to test it failed because of a defect in the test, and we recount that failure here because the episode belongs to §6 and is the reason the current test has the form it has.

Attempt 1, invalid. Classify the 350 states by the position of the designated item in the ordered pool: outside the pool / already selected / in a stratum different from the cut / in the stratum of the cut. The prediction was that the last class would contain exactly the 17 at the ceiling. Result: 25, and, decisively, only 1 of the 17 fell in it. **FRESH_CANDIDATE_POOL** and **GLOBAL_FRESH_PATTERNS** are module-private constants of **brief.ts** (l.111, l.135), and the ordered pool the coverage slots are drawn from is built inside **buildBriefDiverse** (**interleaveFresh** over the two sub-pools, then **pickDedup**, which places the pinned items first and deduplicates as it fills) and returned by no export, so that pool was a reconstruction, and the 24 violations of the prefix assumption were the symptom. Testing a derivation against a reconstructed pipeline does not test the derivation: it tests the reconstruction.

Test 2, on recorded quantities. Proposition 1 has a consequence that does not require assembling any pool:

In every state in which the served set changes, each id that enters must share **last_served** with some id that leaves.

If an id enters from a stratum in which nobody left, the bonus crossed a stratum and Proposition 1 is false. The three quantities (**would_enter**, **would_leave** and **last_served** in the pruned serve-state) are recorded by production and by the replay, not derived. Caveat. “350/350” is coverage, not validation: it means that the 350 brief states of the window were reproduced, each from its own recorded serve-state. What was *validated*, and against what, is in this section (§5.6): the pool reconstruction was withdrawn for producing 24 violations of the prefix assumption, and what remains uses only recorded quantities.

replay-oportunidade.mjs --modo porque --corte rowid --so-ts-file ts-350.txt aborts if there is a single entry without a partner. (**ts-350.txt** is the list of the 350 state timestamps. It was not kept and its hash was not recorded. The same 350 timestamps are the **ts** fields of **out/c-350-v3.json**; that they match the original input byte for byte is not verified.)

Result.

states replayed	350 (errors: 0)
-----------------	-----------------

states in which the served set changes	17
ids that enter, summed	20
entries with no exit in the same stratum	0
Proposition 1	survives

The 17 and the 20 exactly reproduce the independent dose artifact (`mexeu = 17`, `churn_total = 20` at `w = 100,000`), and that reproduction is what makes the zero interpretable: it guarantees that the two runs speak about the same population.

Caveat. The zero was also checked against the possibility of being a test that does not look. We mutated the `stratum` function to give each id its own stratum, under which no entry would have a partner; the mode then reports 20 violations in 17 states and aborts. The assertion bites at the maximum possible, so the 0 in the table is a result, not instrument silence.

Provenance of the run. Corpus = epoch snapshot `e20260826T060003Z.db` (`sha256` of its first MB `56826f69...`), probe exclusion = none, cut by insertion order, designation with `sha256` verified. This provenance is identical to the dose artifact’s. Caveat. The first execution of this test diverged (13 states, not 17) because I had changed two things at once (live corpus and exclusion of the health probes: every preserved probe list has 5 `brief_ids`, but this first run was not preserved, so how many it excluded cannot be checked); the diff of the `procedencia` block of the two artifacts showed both. Diffing provenance before comparing a number is a rule of this work.

Caveat. That corpus no longer exists: epoch retention pruned it, and it is not among the four preserved trial databases nor on either host or the author’s machine (checked 2026-10-03/04). The run can be re-read from its artifacts but not re-executed on its original input.

5.7 The 4.86% ceiling is a property of the **FORMAT**, not of the comparator

Proposition 1 says that the bonus permutes within a `last_served` tie and never crosses it. From this it follows, deductively, that the reachability ceiling is not a property of the comparator alone: it is a property of how many ties the `last_served` format produces. The resolution of `served_at` is inherited from SQLite’s `datetime('now')`: seconds. Nobody chose it as a design parameter; it is the default of a library function.

The counterfactual that separates the two things is the same replay with the stratum key truncated and padded back with zeros (`--granularidade`, in `replay-oportunidade.mjs`). Truncating is surgical here because of a fact verified in the source: `served_at` has a single live consumer in serving, the stratum key. The other (`serveCounts`, the window of the novelty penalty) is exported and tested, but no production path calls it; it is the remnant of mechanism A, replaced by coverage in the 2026-06-26 tune (`brief.ts:588`).

resolution of <code>served_at</code>	briefs that change	ceiling
second — production’s	17 / 350	4.86%
minute	127 / 350	36.29%
hour	281 / 350	80.29%
day	348 / 350	99.43%

Corpus, designation, cut, dose and the set of 350 states are byte for byte the same in the four rows; the consolidating script aborts if any of them diverges, and the native granularity must reproduce the published 17/350 before the table is emitted (both assertions were tested by mutation). Artifact: `CEILING-GRANULARITY-2026-08-28.json`.

The intuition for why lies in the size of the ties, and it comes from another population: the 1,787 chunks that were ever served, over the whole history of the `brief_log`, without the per-brief pruning that the replay applies. In that population, truncating takes the number of distinct strata (counts not preserved in any artifact) from 1,139 (second) to 951, 311 and 57, and the largest stratum from 14 to 61, 89 and 186. The two sets of numbers answer different questions and must not be read on the same line: one describes the tie structure of the served corpus, the other counts briefs that change in the replay.

The reading, and the care it demands. A mechanism that reaches 4.86% of briefs would reach 36% if the system stored the time with one field fewer.

Caveat. It would be overclaiming to say that the ceiling is “a fact about the width of a text field”. It is a fact about the ratio between serving cadence and key resolution: the serves concentrated in the same second create the ties, and in a system that served one item per hour the same field width would move almost nothing. What the measurement establishes is that one of the two terms of that ratio was inherited from a default (the resolution comes from SQLite’s `datetime('now')`, not from a design decision), and that changing only it shifts the reach of the mechanism by an order of magnitude. We did not isolate the two terms; doing so would require repeating the table over a corpus with artificially spaced cadence.

5.7.1 And the ceiling also depends on WHICH chunk the draw picked

The designation rule chooses one chunk per signature group by a draw with a declared seed. The choice within the group is, by construction, arbitrary, so the 17/350 can be a property of the comparator or an accident of the draw, and the difference matters for how the number is read.

We redid the same replay with eight alternative designations, generated by the same rule over the same population, varying only the seed. The seeds come from a family derived deterministically from a fixed phrase in the script (choosing them by hand would allow fishing for the result, since the beacon is public), and all eight are reported (`CEILING-DESIGNATION-SENSITIVITY-2026-08-28.json`):

	change / 350	ceiling
designation in force	17	4.86%
minimum of the eight alternatives	17	4.86%
median	20	5.71%
maximum	26	7.43%

The arbitrariness is bounded by construction: 12 of the 19 groups are singletons, so no seed can move more than 7 designated items.

Two readings. The first, which the data support: the ceiling is robust in order of magnitude: nine draws of the same rule give from 4.86% to 7.43%, and none comes close to changing the conclusion that the mechanism reaches a few percent of briefs. The second: the designation in force is at the minimum of the distribution, tied with `sens-03`, which also gives 17, and not alone. Reporting 4.86% as “the ceiling of the mechanism” describes the extreme, not the rule.

Caveat. Even so, one must not overstate. With 8 alternatives, two of the nine results share the minimum (the designation in force and `sens-03`, both 17), so under exchangeable permutation the probability that the pre-specified designation sits at the minimum is $2/9 \approx 22\%$: being at the minimum does not establish that it is atypical. The median 5.71% is an $n = 8$ point, with deviation ≈ 2.9 states; any reasonable interval covers 17 to 22, so the median should not be read as “the value the rule produces”. What is supported is simpler and weaker: where this paper says 4.86%, read *the ceiling under the designation in force*, and the ceiling of the rule is a distribution that we did not measure precisely enough to summarize in one number.

Caveat. The alternatives also do not move only *more* briefs: they move different ones. The intersection with the 17 original states stays between 9 and 14 (`estados_em_comum_com_a_publicada` in the artifact; it concerns replay states, not which designated items were drawn). The granularity finding, in which the beneficiary changes identity when the resolution changes, shows the same pattern: which chunk receives exposure is decided by details that nobody chose as policy.

Caveat. We did not measure the interaction between the two axes (designation $\times$ granularity); the table of §5.7 is entirely under the designation in force, and this one is entirely under second resolution.

Caveat. A growing part of this table is decided by a tie-break that nobody declared. Truncating creates ties, and when two candidates tie on `last_served` and on `salience`, the comparator does not separate them; the sort is stable, so what decides is the order in which SQLite returned the rows. The SQL pre-rank of this channel orders by a coarse expression (`0.55·importance + 0.10·pain + 0.1 if access > 0`) that takes six

values over the 108; the comparator itself re-sorts by `calculateSalience` (§4.3.2), which takes 15 values over the same 108. An earlier version of this table counted ties on the pre-rank expression (`measurement/empates-por-granularidade.py`, `TIEBREAK-EXPOSURE-2026-08-29.json`: 68 / 269 / 917 / 1,656 pairs), a coarser key than the comparator’s. Exposure measured on the comparator’s key, at three instants inside the 350-state replay window that the input covers (2026-08-26 20:37Z, 22:00Z and 23:52:09Z; `measurement/sprint-empates-salience-producao.mjs`, `out/TIEBREAK-EXPOSURE-PROD-2026-10-05.json`), with the pre-rank expression at the same instants for comparison:

resolution	indistinguishable pairs	% of pairs	largest block	pre-rank expression, same instants
second — production’s	34–42	0.59–0.73%	4	45–59
minute	212–238	3.67–4.12%	10–12	277–322
hour	472–749	8.17–12.96%	18–33	663–1,024
day	1,197	20.72%	42	1,656

The input is a read-only copy of the 2026-09-07 trial epoch (the one behind `out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json`), with `importance`, `pain` and `access` fields as stored there and each candidate’s `last_served` taken from the `brief_log` rows up to the instant. The pool at each instant is exactly the 108 ids of §4.3.1. The day row of the pre-rank column reproduces the published 1,656; the published second, minute and hour values come from an unrecorded instant of 2026-08-29 that this input does not reach. The ties that remain under `calculateSalience` are chunks of one `lessons.md` ingestion with identical fields, or entity chunks with the same importance, pain, access count and access instant (`out/TIEBREAK-EXPOSURE-PROD-SHARED-2026-10-05.json`, the same run with `last_accessed_at`, `created_at` and a source-file class added to each tie group; it holds at all five instants).

The natural counter-argument, that “the two arms share the same arbitrary order, so it cancels out”, does not hold: `churn` is the symmetric difference of two sets, and permuting the order moves both in an uncorrelated way precisely at the boundary of the cut, where churn is born and dies.

What survives and what does not: the direction and the order of magnitude of the effect, because on the comparator’s key the exposure grows 28–35× from second to day while the ceiling grows 20×, and because at the resolution this paper reports it is 0.59–0.73% of pairs. The exact values of the coarse rows (127, 281, 348) carry an arbitrary component that we did not quantify; doing so would require rerunning the 4×350 with a third, explicit tie-break coordinate, and we declare that we did not do so.

Caveat. A prediction of ours died in this test, and it stays. The design note said: “coarsening only merges strata, never splits them $\Rightarrow$ the ceiling is monotonically non-decreasing, and that monotonicity is the self-test of the instrument”. The count does in fact rise, but the sets are not nested: 1 state leaves from second→minute, 2 from minute→hour. The error is that merging strata also moves the control arm, and churn is the difference between the two arms. As measured, the three lost states split into opposite mechanisms:

- **redundancy** — under minute, the designated item comes to enter the control on its own; the intervention is left with nothing to do;
- **unreachability** — under hour, the designated item’s whole stratum drops below the selection cut, and the bonus does not cross strata. It is Proposition 1 biting in the opposite direction.

So the monotonicity of the count is empirical, not structural, and the consolidator reports it instead of requiring it; a guard that asserted nesting would be wrong and would have hidden the finding. And the effect is not only one of quantity: in one of the states the id that enters changes from 308284 under second to 308296 under minute. The resolution of the timestamp also decides which chunk benefits.

5.7.2 A third axis: the anchor measured, the ceiling not

This finding comes from an audit of the artifact chain (2026-08-30), not from review. The replay that produces the 4.86% ceiling ran with `sondas_excluidas`: `[]`: none of our own health probes was removed from the `serve_state`. There existed, in the repository, two artifacts recorded on 2026-08-27 and never read

by anyone (`out/ancora-sondas.json` and `ancora-sem-exclusao.json`) which show that this choice is not neutral:

	without excluding	excluding the 25 rows of 5 probes
distinct <code>last_served</code> groups	44	43
position of the first chunk of the study	3	0

The anchor moves, and the paper reported “two axes” when there are three. The ceiling moves with it, as measured on 2026-08-30, after this section had existed for two days stating that we did not know:

	without excluding	excluding the probes
states that change (of 350)	17	13
ceiling	4.86%	3.71%
total churn	20	13
sensitive states in common	—	1

Two readings, and the second is the stronger.

The ceiling falls by about a quarter. The published 4.86% includes the effect of our own verification probes: five briefs that exist because we *confirmed that the mechanism had gone up*. The act of measuring added itself to what was being measured, and the decontaminated version of the reachability is smaller than the published one. The direction matters for how the paper is read: the published ceiling is a loose upper bound, and loosened by us. The argument that the surface is narrow comes out reinforced, not weakened.

Identity is hardly preserved: of 17 and 13 sensitive states, only ONE is common to both arms. The sets are not nested. The exclusion does not remove some states from a stable set; it almost entirely reorganizes which set is sensitive. This is a much larger sensitivity than the difference between 17 and 13 suggests, and it only appears when identities are compared; anyone looking only at the totals would read “four states fewer” and conclude that the convention hardly matters.

How this was measured, given that the original condition was lost. Paired design: the two arms run over the same corpus of 2026-08-30, the same log of 350 states, the same designation, and vary only the exclusion; the provenance of the two artifacts differs in exactly two fields, `gerado_em` (2 seconds) and `sondas_excluidas`. The 350 states are identical and in the same order, with the same `rowid_corte`, verified element by element.

Caveat. The paired arms ran on `e20260830T060001Z.db` (sha256 of its first MB 3679bca8...), an epoch snapshot under the same retention that pruned `e20260826T060003Z.db` (§5.6). It is not among the preserved trial databases, and it was not found on the research host or on the author’s machine on 2026-10-03; the production host was not searched. Both artifacts survive. The input that would let a third party re-run them is, as far as we know, gone.

Caveat. The control that makes the comparison with the published number valid, at the level of identity rather than count. The arm without exclusion, run on the 2026-08-30 corpus, returns $17/350 = 4.86\%$, the value published on the 2026-08-26 corpus. A matching total alone would be the kind of evidence this section has just disqualified (17 and 13 sensitive states share only one), so the comparison was redone state by state over the existing artifacts (`measurement/sprint-comparabilidade-identidade-572.py`, `COMPARABILITY-IDENTITY-5.7.2.json`). The 350 state keys (`ts`, `agent`, `rowid_corte`) coincide 350/350. The sensitive set is the same 17 states, not merely 17 states. In each of the 350 states the ids that would enter and leave are identical. The undosed control arm (the top-10 served at $w = 0$, which depends on corpus and serve-state but not on the dose) is identical, in order, in 350/350 states (`out/c-350-v3.json` on 2026-08-26 against the 2026-08-30 arm). Changing the corpus moves neither the count, nor which states are sensitive, nor the undosed ranking, so the arm with exclusion is comparable to the published one, and not only to its own pair. The check can see a difference when one exists: applied to the probe-exclusion arm it keeps 1 of 17 sensitive states and 1 of 350 control lists, and a mutation that keeps the count at 17 but changes one entering id makes it fail. Without this control, the difference would be attributable to the corpus as much as to the probes. It

does not recover the level of the ceiling under exclusion on the original corpus, which remains unrecoverable (below).

Note. The artifacts are `out/CEILING-PROBE-EXCLUSION-none-2026-08-30.json` and `out/CEILING-PROBE-EXCLUSION-probes-2026-08-30.json`.

Caveat. The original condition is no longer reproducible, but the sensitivity is measurable. These are two different things ($\leftrightarrow$ F-3.1). The replay corpus, `e20260826T060003Z.db`, was an epoch snapshot and was rotated; on 2026-08-30 only those from 2026-08-28 to 2026-08-30 remained, and the daily backup of 2026-08-26 is from another instant: the `sha256` of the first MB does not match the published pin. So the level of the ceiling under probe exclusion is not recoverable.

What was recoverable is whether the exclusion moves the ceiling, through the paired design described above, which is the same design as that of the two anchor artifacts, one level down. Two conditions made this possible, and were verified before running:

- the closed-window log with the 350 states survived the rotation (it is an `.ndjson`, not a database snapshot), and the designation on disk matches the published `sha256`;
- sweeping the 2026-08-30 corpus by the same signature that identified the probes on 2026-08-27 (a brief with 5 rows instead of the 10 of an organic brief), exactly the same five appear, and no new one. The exclusion is the same act in both corpora.

A pair of artifacts seemed to answer this, and does not. On 2026-08-30 the serving machine held a `campo-churn.json` and a `campo-churn-sem-exclusao.json` (not preserved in the repository or the trial’s evidence archive; their current existence is not verified), and a direct reading suggests a dramatic finding: the fidelity of the replay to the field falls from 0.909 to 0 when the probes are excluded. This pair is not usable. The arm with exclusion was generated at 15:17:11 and does not record `corte_serve_state`; the one without exclusion was generated at 15:18:55 and records `estrito`. The field came into existence in the 104 seconds between the two (by the generation times above; neither file was preserved, so those times cannot be re-checked): they are different versions of the script, and the drop may be due to the exclusion or to the code change, with no way to separate them.

Caveat. Two arms generated minutes apart look paired and may not be. What disambiguates them is not proximity in time, but field-by-field provenance. Here, a field *absent* on one side was the only sign that the instrument had changed midway. A missing field does not stand out on reading; it took diffing the two provenances to see it.

Note. What this changes in the reading of the ceiling: the 4.86% remains the value under the convention declared in the artifacts, and the two measured axes remain valid. What falls is the suggestion that the measured axes exhaust the sensitivity. They do not, and the third, once measured, is the largest of the three in relative effect on the identity of the sensitive states, even though the smallest in effect on the count.

Caveat. The process lesson, which is bigger than the number. This axis existed in two artifacts recorded on 2026-08-27 and never read; it was found by mechanical audit of the script→artifact→claim chain, not by review. Between the discovery and the measurement, this section spent two days stating “*whether the ceiling moves with it, we do not know*”, an honest declaration of ignorance, and an unnecessary one: the measurement cost two runs of half an hour over material that was all on disk. Declaring ignorance is cheaper than measuring, and that is why it is the path taken by inertia. Before writing that one does not know, one should ask how much it would cost to know.

6. Instrument defects: what an auditor of these numbers needs

This section serves contribution (iv), the executable diagnostic: whoever reproduces the measurement needs the defects that changed its numbers. The catalog itself is not a declared contribution (§1). The full catalog has 17 entries and lives in Appendix E. Here are the eight that changed a number this paper reports, because without them the reader has no way to audit the numbers. That is the cut criterion, not the interest of the lesson.

Note. The criterion excludes one that would be the most quotable of all: an aggregate κ of 0.874 reported alongside a stratification that depends on a different boundary, where the panel agrees only 0.31–0.53. It does not enter because it does not change any number in this paper; it changes the analysis stratum of the interventional study, where the split is reported; the aggregate 0.874 itself is recorded in Appendix C.

defect	number it changed
positive control run on a reimplemented pipeline	produced “absurd dose $\Rightarrow$ zero effect”, which became the central retraction. The real pipeline gives 20 events
test of a derivation on a RECONSTRUCTED pool (the coverage pool is not returned by any exported function)	classified 25 states as alterable against 17 real ones, and only 1 of the 17 fell in the class; 24 violations of the prefix assumption were the symptom. The valid test uses only recorded quantities (§5.6)
served_at with second resolution and 6 agents firing within 1–2 s	46.9% of briefs share the second; no temporal cut reproduces the state. Under a strict cut the replay invents 3 and misses 1 event: the outcome comes out 14 instead of 12 (+16.7%)
coarse grid	saturation <i>seemed</i> to fall exactly at the top of the registered band; with 23 doses it is in (4.0, 4.4]
per-chunk search telemetry silent from 2026-05-19 14:47:04, at a deploy boundary (a 1 h 19 min gap in the rows, then null until the writer was restored on 2026-08-27 22:18Z, commit 32f78109), without a CUT — this project’s convention for deliberate withdrawal	comparison across surfaces within a window is impossible, and went 3.3 months without anyone noticing
window of one sub-pool applied to a batch of the other	§4.3.1 said each batch feeds the coverage channel for 7 days and expires. There are two sub-pools: per-agent (sessions/% , 7 d) and global (entities/% + lessons.md , 30 d). The retrodiction batch belonged to the first, the prediction batch to the second. The dated prediction refuted the section, which was replaced: the eligible pool is 108 chunks (0.16% of the corpus) and is 100% exhausted every day
brief_log does not record the channel that served each row	counting serves of a batch measures the UNION of coverage with the main pool, and the main pool has no age filter. It made a guard flag “age 7.42 served” as a violation of the 7-day window — serves that were never subject to any window. Eligibility excludes but does not attribute; the attribution is the paired replay of the same state with freshSlots = 2 and freshSlots = 0 (§4.3.1, CHANNEL-ATTRIBUTION-2026-08-29.json)
comparison of a filtered count with an unfiltered one	inverted the sign of a conclusion: 617×245 cumulative becomes 245×≥151 in the common window

The pattern running through all eight, and the transferable finding, is that each one passed verification and survived. In each, the verification instrument shared the wrong premise of what it verified. A positive control run on the reimplemented pipeline confirms the reimplemented pipeline; a census of dead columns done by **grep** inherits the blindness of **grep**. The defense that worked, in all eight, was the same: reproduce a published anchor before varying anything.

Note. There is also an asymmetry: four of the eight were found because a number matched too well: saturation exactly at the top of the registered band, an absurd dose with exactly zero effect. Suspicious agreement was a better detector than disagreement.

6.1 How much of this paper the verifier actually verifies

A verification discipline that does not measure itself is a statement of intent. **claims_check.py** recomputes claims against an artifact and fails if they diverge, but until 2026-08-29 nobody had asked *over how many*. Adversarial review asked the question, and the first measurement (**measurement/censo-de-alegacoes-sem-guarda.py**) showed 19 of the 32 curated numeric claims with no guard at all (59.4%).

And the artifact of that measurement no longer exists, for the very reason this paragraph describes. **out/CLAIM-COVERAGE-2026-08-29.json** is rewritten on every run of the census: as of 2026-10-03 it still reads 10 of 32 (31.2%), which is the state after the 2026-08-29 round, not the 19 that the text attributed to it. An earlier version of this sentence cited it as the source of the 59.4%, and the citation had been false since the second time the script ran. The number 19 has as its record this paragraph and the round’s commit, not an artifact, which is exactly the defect the section catalogs, committed by the section that catalogs it.

Found by adversarial review (Codex, 2026-08-30); the operational lesson is that an artifact rewritten in place does not serve as a historical citation: either the name carries the instant, or the claim cannot point to it.

There were, therefore, two rounds of guards, and the earlier text mentioned only one: the one of 2026-08-29 closed nine (19 $\rightarrow$ 10) and the one of 2026-08-30 closed the remaining ten unguarded claims, reaching 0 of 32 (0.0%), and strengthened five weakly guarded ones. The table lists all fifteen; the five that were weakly guarded rather than unguarded are in italics (`out/CLAIM-COVERAGE-2026-08-29.json`, the state after the 2026-08-29 round, has exactly ten `SEM_GUARDA`). They came from three distinct origins, and the distinction matters because only the third was serious:

origin	claims	remedy
derivable from numbers already locked	99.98% $\cdot$ <i>8.7$\times$</i> $\cdot$ <i>brief coverage</i> $\cdot$ 46,280 $\cdot$ 82.2%	recompute in the guard
existing artifact, never read	0.161% $\cdot$ 12.4 $\cdot$ 4.86% $\cdot$ 36% $\cdot$ 80% $\cdot$ <i>32.1% $\cdot$ 232</i>	open and compare
artifact that did not exist	−0.961 $\cdot$ −0.728 $\cdot$ 0.471	the script wrote nothing

The third row only appeared when writing the guard, and it is the finding of this section. `measurement/robustez-tamanho-exposicao.py` ran from 2026-08-27 to 2026-08-30 without writing any artifact (no `--out`, no `json.dump`). The three coefficients that support the whole of §4.2, including the β that is the result that *survives* the 15 types, existed only on `stdout`, were read by a human and transcribed by hand into the text. Changing the input CSV would change all three and the manuscript would go on asserting the old ones, with nothing to flag it. It is the exact form of the defect this section describes (prose asserting a computed result is a cache without invalidation), found in the numbers the paper uses to defend itself against the strongest objection it received.

Note. The gap sat in what §4.2 presents as the most robust estimator, not in a peripheral number. The degree of protection of a claim bears no relation to its centrality; if it did, this one would have been the first. `out/SIZE-ROBUSTNESS-2026-08-30.json` is the first artifact the three have.

This gap had already produced an error, which only appeared because the gap was measured: the manuscript asserted 583,973 cumulative slots in five places. The artifact says 583,763. Two digits and 210 slots of difference, no artifact with the value in the text, and nothing to flag it. It was the most-cited number in the paper, the basis of the capacity thesis, and the least protected. In addition, the field is called `slots_historicos_ATE_AGORA_serie_viva` and the quantity was growing by about 6,600 per day at the time (the closed week’s 46,295 slots / 7 = 6,614, `slots_7d` in `out/superficie.json`; the series stood at 583,763 on 2026-08-28 09:52Z in the same artifact, and a reading of 591,323 on 2026-08-30 has no preserved artifact); and the sentence said “*today*”, which that rate refutes: at about 6,600/day, three weeks later the value would be about 722 thousand. Corrected on 2026-09-21; the instance was in this same paragraph that states the rule. Citing a live series without fixing the instant is writing a number that ages into falsehood on its own; for that reason the claim carries the 84.7 days and the artifact’s `T_REF`, and the new guard recomputes the derived values (8.7 $\times$ and 2.43%) instead of only checking that the text did not change.

And recomputing was not enough. The recomputed value was 1,787 / 67,187 = 2.66%, which mixes a historical numerator (it includes the 152 deleted later) with a live denominator, the defect that F-3.2 corrected in the 83.78% and that the rule in §4.1.1 (“*the cited percentage is over the live corpus*”) prohibits. The guard faithfully recomputed the wrong calculation. Corrected on 2026-09-21 to 1,635 / 67,187 = 2.43%, by adversarial review.

The census of universes did not catch it either. `censo-de-universos-no-paragrafo.py` exists for this family and has 2,66 in its table, labeled (“cobertura do brief”, “brief”). It compares the declared universes of adjacent numbers, and a number has only one label: the mixture that lives inside a single quotient is invisible to it by construction. A guard that compares labels does not see the calculation that the label summarizes.

And the occurrence count became part of the guard, because `valor in texto` is satisfied by any occurrence: three mutations passed the first test because I changed only the first of several mentions of the same number.

The assumed cost is that a legitimate edit that changes the number of mentions has to update the count on purpose.

6.2 Pinning an artifact’s identifier does not preserve it: two losses and one mis-scoped search

The eight defects above are instrument defects: a number measured wrongly, a constant with no referent, a guard that does not bite. This one is of a different nature, and only became visible when what we took to be the third occurrence appeared: three reports in three days, from three independent mechanisms. Rechecked on 2026-10-03, the third was not a loss of the pinned artifact (third row):

artifact	how it was pinned	what happened
corpus of the ceiling replay (e20260826T060003Z.db)	sha256 of the first MB, published	rotated — epoch snapshot, retention deleted it; the same day’s daily backup is from a different instant and does not match a merge rewrote the hashes; recovered on 2026-08-30 and re-pinned by sha256 of the bytes
three <i>servicing</i> modules (brief-diversity, salience, search)	commit hash	searched on 2026-08-30 by name, size and hash on the servicing machine only , and reported missing. It is not: on 2026-10-03 it was at its declared path on the other host, with sha256 equal to the pin (ba5fcc81...) and its 3,860-line manifest intact. The 280 adjudicated episodes (p2_verdict, adjudicated 2026-08-15 → 08-21) were never in it (0 of 280 ids among its 5,547 episodes; id list in _sprint-2026-10-04/A-aging/p2_verdict_ids-280.txt), and none of them survives in the live archive or in the preserved episode files of the trial
frozen corpus of the interventional study (action-archive-20260729T094609Z.tar.gz)	sha256, path and mode 0400 declared	

All three pins were correct; the first two failed in the same way, and the third did not fail. A published sha256 identifies an artifact without requiring trust in whoever published it, which is exactly the property one wants. What it does not do is prevent the artifact from disappearing, and these two things are easy to confuse because the hash *looks like* an act of preservation. It is an act of identification. Retention does not read hashes.

The third case shows the cost in its pure form, though not in the way we first wrote it. The panel adjudicated 280 episodes between 2026-08-15 and 08-21. The table kept **sig_primary**, **severity** and **chunk_id**, and the judged material is gone: 0 of the 280 ids appear in the frozen July corpus or in the preserved episode files of the trial (checked 2026-10-03). What remains is a verdict without evidence: verifiable against itself and unrecoverable for any third party. The frozen corpus itself was never lost. Our search looked at one host and reported “does not exist”; the file was at its declared path on the other host, byte-identical to the pin. That search was a guard whose predicate lacked the data, the same class this section catalogues. And the **sig()** function, versioned in a repository rather than pinned by hash, survived as expected: its sha256 (e860357bd9f1fc06...) matches the frozen one because the file is in git.

The rule that comes out of this, and that holds for any deposit: deposit the blob, not the hash. If the artifact is too large to deposit, then the reproducibility claim that rests on it has to say that it is perishable. That is a considerably weaker claim, and it is the true one.

Note. None of the three was found by review. The first two came from a mechanical audit of the script→artifact→claim chain; the third, from trying to use the corpus and not finding it on the host we searched. A reviewer reading **CORPUS-FREEZE.md** sees an impeccable document (path, size, hash, file mode, verification command), and the file is where it says. Our own search still declared it missing, because it covered one host. No reading distinguishes a valid pin from an orphaned one, and neither does a search scoped too narrowly; only retrieving the artifact does.

7. Threats to validity

- **n = 1 system.** This is the main threat and has no mitigation within this paper. The generalization of the mechanism is deductive (§5), not empirical: it holds for any ranker with lexicographic order and a bonus on the subordinate coordinate that serves a prefix of that order, and whether other systems have this structure is an open question. Partial mitigation: publishing the executable diagnostic so that third parties can measure their own system;
- small types (**decision** n=11, **person** n=14, **feedback** n=17) do not support an individual reading; they enter only in the correlation test;
- **access_count** is “a top candidate of at least one tracked search sub-query”, whoever initiated the call and whether or not the chunk was returned (§3.1), with no per-event history, and not “exposed at least once”: the brief never writes to that column (§2, §4.3.2). Whoever measures exposure needs the union with **brief_log**, never this column alone;
- ~~the per-day coverage diversity has a regime break on 2026-08-21-22, still without explanation~~ → explained (§4.3.1), and the first explanation we gave was refuted by a dated prediction before it reached the deposit (§6). It is not a regime break: the eligible pool has 108 chunks and is 100% exhausted per day, so daily diversity measures when new material entered within the patterns, not rotation. It becomes a declared limit: an outcome built on daily diversity is non-stationary because of its dependence on the ingestion calendar, and a window with no new ingestion measures zero by construction;
- in the windows from which this paper’s rates and replay results are computed (the closed brief week [2026-08-20, 2026-08-27), the census of 2026-08-28, the 350 replay states of 2026-08-26/27 and the pool measurements of 2026-08-26 to 08-29), the intervention ran in shadow mode: the counterfactual is observed and nothing treated was served, so every rate here is an opportunity rate, not an effect. The serving-log set comparisons of §4.3.1 extend to 2026-09-21 and so span the trial’s control and treatment epochs; they are descriptive. From 2026-09-01 10:25:39Z to 2026-09-21 09:43:05Z the separately registered trial did serve treated briefs (11 treatment epochs out of 19 with data; **DEVIATIONS-FOR-PAPER.md** §10.29, §10.31). Its results belong to Paper B, not to this paper.

8. Related Work

8.1 What the agent-memory field measures

The canonical survey (TMLR 2602.06052v4) partitions the metrics in use into three families (accuracy-based, similarity-based, and LLM-as-judge), and all three score the representation: given a set of queries, how well the system retrieves what is relevant. MemoryArena (2602.16313) and Evo-Memory (2511.20857) are the closest neighbors, and both compare systems on fixed tasks. Neither measures what the system delivered to the agent under the traffic it received.

The empty cell matters because a retrieval metric is conditional on a query having been issued. An item that no query reaches and that no brief includes has an undefined nDCG, not a low one. That is the population this paper measures, within a stated scope: 56,288 live chunks (83.78%) have neither a brief-log record nor a positive search counter, a lower bound on non-delivery by the brief and by tracked search; non-delivery across all agent-facing search is not established (§3.1).

8.2 Exposure in recommendation: where the vocabulary exists

The notion that delivered attention is a finite, allocable resource, distinct from estimated relevance, is mature in retrieval and recommendation. Singh and Joachims (KDD ’18) formulate exposure allocation as an optimizable constraint; Diaz et al. (CIKM ’20) make expected exposure an evaluation metric, over stochastic rankings. The *popularity bias* literature measures the correlated phenomenon on the catalog side: long-tail items receive disproportionately less exposure than their prevalence (survey by Klimashevskaya et al., UMUI 2024, covering 123 works). Chaney et al. (RecSys ’18) show that the feedback loop between what is exposed and what is learned increases homogeneity over time.

Our result is *popularity bias* where “popularity” is the size of the collection to which the item belongs.

Chaney’s loop has only a partial analogue here, and the difference is one of design, not chance: `access_count` enters `salience`, but is incremented only by search (`search.ts:396`); the brief is declaredly *read-only* over `chunks`. So serving in the brief does not raise the priority of anything, and the classic loop, in which exposure reinforces itself, does not close inside the system (§4.3.2). What exists is a permanent, non-decaying encoding of past tracked search traffic, whoever initiated it (§3.1). If the loop closes, it closes through the agent, who sees the item and perhaps searches for it again. We did not measure that.

The work closest to our finding is Bower et al. (2022), which shows that exposure inequality originates before the ordering, in the candidate set, and that randomizing the next step can even worsen it. We agree on the location and diverge on the mechanism:

That literature presupposes that exposure responds to the score being controlled, which is what licenses redistributing exposure by making the ranking stochastic. Here exposure stays weakly monotone in the score (§4.4: monotone in each of the 350 states) but responds only within ties of the dominant coordinate: the coverage slots are ordered by a lexicographic comparator, the score is the subordinate coordinate, and an additive bonus on the score acts only within those ties (§5.2). A lever built on the score does not move what the dominant coordinate has already decided. This follows from the algebra of the comparator, not from the bonus’s magnitude.

Hence the bridge is not merely terminological. The vocabulary of surface capacity matters for agent memory, and the standard redistribution technique does not transfer without first checking on which coordinate the ordering decides. That check is the diagnosis we publish.

Two scope caveats. The analogy is one of mechanism, not of application: there the consumer is a human user with decreasing position and attention; here it is an agent that receives 10 items at once, and there is no position model. And “exposure fairness” is a normative question that we do not raise: the argument here is one of utility and diagnosis, not of equity among items.

8.3 Pre-registration in systems CS

Prospective registration of hypothesis, outcome, and analysis is routine in clinical trials and in parts of psychology; in systems CS, it is not. The survey of 218 papers from §8.1 has zero occurrences of any spelling of *pre-registration*; measured together, it also has zero of `randomized/randomised` and of `ablation` (§1). The absence is of the entire methodological family, not of one term.

This paper does not claim to be that precedent. The prospective registration we deposited (OSF yf7d2) is for another study: a randomized crossover on the behavior of the agent. Corrected on 2026-09-21: an earlier version of this sentence said it “*did not run*”. It did: from 2026-09-01 10:25:39Z until the dose was switched off on 2026-09-21 09:43:05Z (the window closed at the 2026-09-20 epoch; 20 epochs realized, 19 with data). Its results are Paper B, not this one. What remains true, and what the sentence meant to say, is that this manuscript is not that study: what it reports is descriptive and was not pre-registered.

Correction. The claim remained false for three weeks in a paragraph whose function is to correct the public record. A reviewer who opened OSF yf7d2, saw the dates, and cross-checked them against this line would disqualify the paper’s entire self-audit thesis with a single counterfactual. It was found by adversarial review, not by the sweep that item 5 of the list *Open items* (at the end of this manuscript) declares open. That is evidence that declaring a sweep open is not the same as performing it.

Appendix A records the deviations from that registration anyway, and the reason is narrow: as long as the public deposit exists asserting things that the measurement contradicts (including two that understate the design itself), leaving them standing is choosing to let the error survive. This is an obligation of correction, not a methodological credential.

What the absence of experimental vocabulary in the survey supports is more modest: there is no established convention on what to declare before intervening in a live memory system, and we found that out the expensive way.

Provenance of this section. MemoryArena and Evo-Memory were read in full on 2026-08-15 (RELATED-WORK.md §4 and §4.1); the survey, in full on 2026-08-13, with the counts recomputed on 2026-08-28 over the

PDF pinned by sha256. The recommendation papers of §8.2 were read in abstract and metadata, not in full text. I declare this because the claim I make about them is about the monotonicity presupposition, and full text could refine that claim.

9. Discussion

Three claims, and none beyond them.

First: this system’s non-exposure is the result of policy, not of capacity. This is the claim that the measurement inverted relative to the hypothesis we started with. What supports it is one line of arithmetic plus one mechanism, not a statistical test: 583,763 slots (the live series frozen at 2026-08-28 09:52:08Z, the `ate` field of `out/superficie.json`, §6.1) against 67,187 chunks, capacity to show everything 8.7 times over, while the brief served 1,635 distinct live chunks (1,787 counting the 152 later deleted; §4.1). There is no physical constraint to remove here; there is an ordering that revisits, and §5 shows by which path.

Note. This section used to support the same conclusion through the `slots/distinct` ratio (printed then as 325; the locked numbers give $583,763/1,787 \approx 327$), withdrawn for the reason given in Appendix F-1.

Caveat. The correct objection to this is that slots are not fungible: the surface delivers 10 per session, and nothing guarantees that a session tolerates more than 10. That is true, and the claim does not depend on it. Showing ten different items per session would never violate the limit of ten; what is missing is rotation across sessions (§4.1.1), not session size. This changes what can be asked for: while the surface looks small, “expose more” is an impossible request; once the slack is measured, it is a request about design.

Caveat. Nor does it follow that the policy is wrong. A 10-item surface has to concentrate, and serving uniformly would destroy its value. What follows is that the boundary between “what the agent sees” and “what exists” was chosen, almost always without anyone choosing it explicitly: `freshSlots = 2` was a configuration default with no override in the measured window (verified on 2026-08-27, `SUPERFICIE-2026-08-27.md`), and the two patterns of `GLOBAL_FRESH_PATTERNS`, together with the importance floor and the 30-day window, carve out 0.16% of the corpus in a way that no design document anticipated.

Second: both channels freeze, and the asymmetry between them is the finding. The coverage channel (2 of the 10 slots) fails by population: 108 eligible in a corpus of 67,187, with zero never-served remaining and 12.4 slots per candidate on a closed day (measured inside the defective-ingestion period of §4.3.1), and by responding to the score only within `last_served` ties, up to a ceiling whose existence is derivable from the code before any experiment and whose value, 4.86% under the conventions in force on the 2026-08-26 corpus (defective-ingestion regime, empty never-served stratum), was measured by replay. The main pool (the other 8) fails for the opposite reason: there the score is the dominant coordinate, and three of its four terms do not decay. The access component is monotone in a counter that only goes up, so the positions of the three constant chunks are determined by tracked search traffic from months ago, whoever initiated it (§3.1; measured counterfactual, §4.3.2: last accessed 90, 30, and 42 days before the window closed on 2026-08-28, in chunks that won 4,632 of 4,632 briefs).

The main pool would respond to a score adjustment, and nobody adjusts it. The coverage channel, designed to compensate for the main pool, has a daily reach bounded by its eligible population (path patterns, the importance/pain floor and age windows), which it exhausted in the measured regime, and within a brief it responds to score only within `last_served` ties (17/350 here) and then saturates. This combination, not capacity, produces the 2.43% (1,635/67,187, live).

Caveat. This differs from the classic recommendation feedback loop: here exposure in the brief does not reinforce itself, since `access_count` is only incremented by search and the brief is declaredly read-only over it. What exists is a permanent, non-decaying encoding of past traffic, not a loop.

We state the design consequence in full: we designed an intervention whose ceiling was derivable from the code before anything was served. Corrected on 2026-09-21: the sentence ended with “*and, at the close of this manuscript, nothing yet has been*”, which stopped being true on 2026-09-01. The trial produced 19 epochs with serving data, 11 of them in a treatment arm. The point survives the correction: the ceiling was derivable from the code before the intervention ran. Whether the served dose changed anything is Paper B’s question,

and its answer is not a measurement of this ceiling. This paper does not claim that the trial left the ceiling unchanged. Whoever is going to intervene on a ranker should read the comparator first and ask *on which coordinate my lever acts*; it costs an afternoon and saves an experimental round.

Third: none of this is visible through the metrics the field uses. This is the only claim that goes beyond this system, because it concerns the instrument, not the result. nDCG and recall are conditional on a query having been issued; an item that no query reaches and no brief includes has an undefined score, not a low one. Measuring the delivery surface requires the system in operation and a per-item record of what was served, and this system almost lacked that record: 16 telemetry columns with no writer during the measurement window (census of 2026-08-27), at six distinct instants, one of them exactly the one that recorded which chunks search returned. Its writer was restored on 2026-08-27 (commit 32f78109), too late for this window (§6).

What we do not claim. That the field optimizes the wrong coordinate: it is one system, and the only generalization we make is deductive. For any ranker with lexicographic order and a bonus on the subordinate coordinate that serves a prefix of that order, the ceiling exists. How many systems have that shape is an open question, and it is to answer it that the diagnosis is published. That the 83.78% without an exposure record is *bad*: part of the corpus is log, and a log does not need to be read to be useful. The number establishes the scale of the population that no retrieval metric reaches, not a harm. And that exposure changes the agent’s behavior: we did not measure it.

Appendix A — Relation to the pre-registration

There is a public pre-registration (OSF yf7d2, Zenodo 10.5281/zenodo.22110203) and this paper is not the study it registers. The registration covers an interventional study (designating chunks, serving a dose, estimating an effect), which went **active** on 2026-09-01 at 10:25:39Z and whose dose was switched off on 2026-09-21 at 09:43:05Z, closing the window at the 2026-09-20 epoch. Of 234 registered epochs, 20 were realized and 19 carry serving data (16 whole and 3 partial under the analysis spec’s clock ruler; 09-02 is empty); 11 were in treatment arms and 8 in control (DEVIATIONS-FOR-PAPER.md §10.29, §10.31). When the first version of this manuscript was closed, the study had not yet begun, and the text below was written under that condition; it no longer holds. The interventional results are not here. They are Paper B (MANUSCRIPT-B.md, a draft opened on 2026-09-21 and not yet deposited), whose underlying records are DEVIATIONS-FOR-PAPER.md §10.29–§10.34, in the version deposited with this text; the v1.0 deposit (Appendix D) predates them, and its DEVIATIONS-FOR-PAPER.md stops at §9.

Caveat. Nothing this paper measures depends on the trial having run: the surface census, the mechanism and the 350-state replay of the ceiling use pre-treatment data (states before 2026-09-01). The longitudinal serving-log checks of §4.3.1 (spanning 2026-08-21 to 2026-09-21; the same 33 main and 108 coverage ids on every measurable day from 2026-08-24 to 2026-09-19) extend into the trial, which served treated briefs in 11 epochs; they are descriptive observations across control and treatment epochs, not pre-treatment measurement. The only thing that changes with the close of the trial is that the sentence “had not begun” can no longer be read in the present tense.

What this paper reports are the measurements made while that study was being built: the surface, the carousel, the mechanism and its ceiling. They were not pre-registered, so they are exploratory and descriptive, not confirmatory.

Caveat. Three things we measured here contradict the registration, and they are stated here because the registration is public and someone will cross-check the two:

the registration states	what was measured	direction
$\Delta_{\text{cut}} = 0.043$ is “ <i>the measured salience spread at the brief cut</i> ”	there is no cut: the code applies no threshold. The comparator is lexicographic and salience only breaks ties between identical last_served values (§5.1)	overstates: the registration promises more

the registration states	what was measured	direction
the {2 · 4 · 7.5} band is among “ <i>what does not move, and could not</i> ”	it moves: 11 / 15 / 17 states out of 350, monotonic, with saturation in (4.0, 4.4] (§5.4)	understates: the registration promises less
(v1.12 §5) the designation is an open defect	closed on 2026-08-26: the seed declaration was pushed at 20:07:24Z, 1,056 s before the drand round it names was issued (20:25:00Z), and the designation was derived from that round (Appendix B)	understates: the registration promises less

The two understating rows are the ones that matter, because nobody corrects on their own an error that favors them: the registration states that a parameter has no effect when it does, and that a defect is open when it has been closed.

Caveat. And the band row changed status because of an instrument defect of ours, not because of new data: the positive control that produced the “*does not move*” was running on a reimplemented pool. That is a matter for §5.6, not a footnote.

The full list of deviations lives in `DEVIATIONS-FOR-PAPER.md` and belongs to the interventional paper. One that affects only the interventional study, the migration of the analysis stratum from S2 to $\geq$ S1 by panel agreement, is not in that log; it is reported in Paper B’s list of deviations (`MANUSCRIPT-B.md`).

Appendix B — Designation chain

Draw with a declared seed: drand quicknet round 31657512 (issued at 20:25:00Z), declaration pushed at 20:07:24Z: 1,056 s of precedence, with the round returning HTTP 425 at the time of writing. An independent reviewer re-derived the designated set using only the public beacon and the deposited CSV.

Appendix C — The panel, and why it is not here

The population of 55 chunks in 19 groups comes from `p2_verdict`, the product of a 3-family severity adjudication panel. The panel, its agreement per boundary and the rate parameter it estimates belong to the interventional study and are reported in Paper B (`MANUSCRIPT-B.md`, Appendix A, relation to the pre-registration), which carries the split agreement: κ 0.87–0.93 at $\geq$ S1 against 0.31–0.53 at the $\geq$ S2 boundary. The aggregate κ of 0.874 that hid this split is not in Paper B nor in `DEVIATIONS-FOR-PAPER.md`; it is recorded here.

For what this paper reports, the panel enters in one way only: it fixed a population. The 19 designated items are a fixed and publicly re-derivable set (Appendix B), and that is all §5 needs. See the caveat in §5.1 on which of the two numbers in §5 depends on the severity label and which does not.

Appendix D — Artifacts

Scripts are in `measurement/`, except `claims_check.py`, which is at the root of `paper2-interventional/`, and the sprint script whose full path the table gives. Artifacts are in `out/` (including the three dated 2026-10-05), in `measurement/` (`CHANNEL-ATTRIBUTION-2026-08-29.json`), at the root of `paper2-interventional/` (`CEILING-*`, `TIEBREAK-*`, `POOL-ELEGIVEL-2026-08-28.json`, `BATCH-CYCLE-*`, `PREDICTION-*`), and, for the ones produced on 2026-10-04, in `_sprint-2026-10-04/` (`POOL-ELEGIVEL-2026-08-26-to-29.json`; `A-recon/RECON-52-e-sondas-2026-10-04.json`; `A-recon/ORGANICO-e-hashes-2026-10-04.txt`; `A-recon-evidence/COMPARABILITY-IDENTITY-5.7.2.json`; `A-filters-disaggregation/out-ord0826.json`; `A-filters-disaggregation/observed-main-from-log.json`; `A-rc2/COVERAGE-SET-FROM-LOG-2026-10-04.json`; `A-aging/WARNING-DENSITY-recomputed-2026-10-03.json`; `A-aging/p2_verdict_ids-280.txt`):

what	script	artifact
exposure surface	superficie-de-exposicao.py	out/superficie.json
replay + dose + threshold + gaps	replay-oportunidade.mjs · replay-resumo.py	out/c-350-v3.json · out/dose-350-v3.json · out/limiar-17.json · out/gaps.json
ceiling granularity (§5.7)	replay-oportunidade.mjs --granularidade · granularidade-do-teto.py	out/gran-{seg,min,hora,dia}.json · out/gran3-{seg,min,hora}.json · CEILING-GRANULARITY-2026-08-28.json
exposure to arbitrary tie-breaking on the comparator’s key (§5.7.1)	sprint-empates-salience-producao.mjs (--shared-plus 1 for the second file)	out/TIEBREAK-EXPOSURE-PROD-2026-10-05.json · out/TIEBREAK-EXPOSURE-PROD-SHARED-2026-10-05.json (what each tie group shares)
the same on the SQL pre-rank expression (superseded as the §5.7.1 table; cited for comparison)	empates-por-granularidade.py	TIEBREAK-EXPOSURE-2026-08-29.json
ceiling sensitivity to the designation (§5.7.1)	sensibilidade-da-designacao.py	out/sens-*.json · CEILING-DESIGNATION-SENSITIVITY-2026-08-28.json
eligible pool of the coverage channel (§4.3.1)	pool-elegivel.py · sprint-pool-elegivel-multidia.py	POOL-ELEGIVEL-2026-08-28.json · POOL-ELEGIVEL-2026-08-26-to-29.json
differential attribution by channel (§4.3.1)	replay-oportunidade.mjs --modo canal	CHANNEL-ATTRIBUTION-2026-08-29.json
served coverage set per day, from the serving log (§4.3.1)	_sprint-2026-10-04/A-rc2/coverage- set-from-log.py	A-rc2/COVERAGE-SET-FROM-LOG-2026-10-04.json
batch cycle, and the refuted prediction (§4.3.1, §6)	ciclo-do-lote.py · regime-cobertura.py	BATCH-CYCLE-2026-08-28.json · BATCH-CYCLE-2026-08-29.json · PREDICTION-2026-08-29.md
exposure to the resolution defect monitoring triggers (retired 2026-09-21; kept as record)	irmaos-no-segundo.py gatilho-saturacao.sh · gatilho-composicao.mjs	— measurement/implantacao/
probe-exclusion axis of the ceiling (§5.7.2)	replay-oportunidade.mjs --excluir-briefs	out/CEILING-PROBE-EXCLUSION- {none,probes}-2026-08-30.json · out/ancora-sondas.json · out/ancora-sem-exclusao.json
identity-level comparability control (§5.7.2)	sprint-comparabilidade-identidade- 572.py	COMPARABILITY-IDENTITY-5.7.2.json
importance-floor composition, cohort exposure (§4.1)	composicao-do-piso.py · exposicao-por-coorte.py	out/FLOOR-COMPOSITION-2026-08-29.json · out/EXPOSURE-BY-COHORT-2026-08-29.json
size axis (§4.2, §6.1)	lacuna-no-eixo-de-tamanho.py · robustez-tamanho-exposicao.py --dados out/tipos-2026-08-28.csv · fig1-capacidade.py	out/tipos-2026-08-28.csv (input) · out/SIZE-AXIS-GAP-2026-08-29.json · out/SIZE-ROBUSTNESS-2026-08-30.json · out/SIZE-EXPOSURE-15-2026-08-29.json; the 13-type Spearman ρ , the three partials, $r(\log n, \text{age})$ and the 9-type r/ρ of §4.2 are printed by the script, not written to the JSON
top-of-pool counterfactual, production salience (§4.3.2)	sprint-contrafactual-salience- producao.mjs	out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json
top-of-pool counterfactual, first version (superseded: not the production function; kept as the anchor)	contrafactual-do-topo.py	out/TOP-COUNTERFACTUAL-2026-08-29.json
bonus at the largest threshold against the largest adjacent gap (§4.4, §5.4)	sprint-bonus-vs-passo.py	out/BONUS-VS-STEP-2026-10-05.json
the 52 deleted chunks, the five probes and the full-pool counterfactual (§4.3, §4.3.1, §4.3.2)	sprint-recon-52-e-sondas.py	A-recon/RECON-52-e-sondas-2026-10-04.json · A-recon/ORGANICO-e-hashes-2026-10-04.txt
main-pool filter disaggregation, and the served main set from the log (§4.3.1)	sprint-desagrega-filtros-pool- principal.py (--log-only for the log census)	A-filters-disaggregation/out- ord0826.json · A-filters-disaggregation/observed- main-from-log.json
warning density recomputed on the 2,058-line text (App. F)	densidade-de-avisos.py	A-aging/WARNING-DENSITY-recomputed- 2026-10-03.json
ids of the 280 adjudicated episodes (§6.2)	—	A-aging/p2_verdict_ids-280.txt

what	script	artifact
verifier coverage and warning density (§6.1, App. F)	claims_check.py · censo-de-alegacoes-sem-guarda.py · censo-de-universos-no-paragrafo.py · densidade-de-avisos.py	out/CLAIM-COVERAGE-2026-08-29.json · out/WARNING-DENSITY-2026-08-30.json
survey count (§1) figures 0–3	survey-string-count.py fig0-arquitetura.py · fig1-capacidade.py · fig2-concentracao.py · fig3-dose-resposta.py	— out/fig{0..3}*.svg
strict-cut field replay (§3.3) deviations from the registration, and the records behind the interventional results (Paper B, not yet deposited) source notes the text cites	replay-resumo.py --campo-estrito —	out/c-350.json DEVIATIONS-FOR-PAPER.md SUPERFICIE-2026-08-27.md · REPLAY-OPORTUNIDADE-2026-08-27.md · REMEDIATION-2026-08-27.md · PROTOCOL-CALIBRATION-2026-08-27.md · CORPUS-FREEZE.md · RELATED-WORK.md · REVISAO-ADVERSARIAL-2026-09-21.md

Deposit. Zenodo, concept DOI 10.5281/zenodo.22181414 (it resolves to the latest version). Version 1.0 (10.5281/zenodo.22181415, published 2026-08-30T21:39:24Z, in Portuguese) holds an earlier text of this manuscript (a 126,564-byte MANUSCRIPT.md, plus 12 supporting files) and an earlier DEVIATIONS-FOR-PAPER.md that stops at §9. It does not contain the §10.29–§10.34 cited in Appendix A, the 2026-09-21 corrections, this English translation, nor the consolidated artifacts of §4.3.1, §5.7 and §5.7.1 listed above (10 of the paths this table listed before 2026-10-04 are absent from it; `_sprint-2026-10-04/A-recon-5.7.2-appD.md`). The version that matches this text is the next one (v1.1) under the same concept DOI. Its version DOI is 10.5281/zenodo.23163119, reserved before deposit. The pre-registration of the interventional study is a separate record (10.5281/zenodo.22110203, v1.12, still the latest version of that record as of 2026-10-03).

Appendix E — Full catalog of instrument defects

The eight of §6 plus the nine below. The separation is by consequence, not by importance: these did not change any number reported in this paper. That does not make them less transferable, and three of them are the lessons I would expect to be the most useful to third parties.

defect	measured consequence
database clock inside the eligibility filter, even though the function receives the instant as an argument the sorter discards the key that ordered the pool monitoring trigger calibrated on adjacent gap	the brief is not a pure function of (corpus, state, nowMs); a naive replay measures another population grouping by it gives one group, silently watches a quantity that does not bound the mechanism: stays green while saturating
attributing the death of an instrument to a commit, without checking the timestamp	I published that 7fdaab4f erased it; the commit is from 2026-05-20 01:03 UTC and the writes stopped ten hours earlier . It removed code that was already mute. A diff that <i>explains</i> the effect is not proof that it <i>caused</i> it
census of “columns without a writer” by grep, by non-null, or by distinct>1	the three give different answers and all are wrong: grep misses UPDATE and dynamic SQL (–3 columns), DEFAULT produces 11 false live ones, and pre-death history inflates the distinct counts (<code>reranker_latency_ms</code> : 394 , all earlier). The valid census is distinct in a later window , cross-checked with the literal INSERT list: 16 columns, at six distinct instants over 10 days — and in one of them a CUT exists, so it is a withdrawal , not a regression
field without a writer used as an origin signature	<code>requesting_agent</code> null was treated by me as proof that the telemetry measured the cron; it is null for everyone since 2026-05-18. The valid test is the cron minute (94.8%)
live series cited as a snapshot	an n changed within minutes and the assertion caught it

defect	measured consequence
value typed inside the verification instrument itself	<code>w_min</code> maximum was pinned at 7.5 in the script and aged into false when the fine grid gave 4.4; it is now a required argument coming from the dose artifact
“the problem is the dissenting panelist” — which here was false	<code>xai</code> is almost a superset of the others (1 of 16 and 1 of 13 outside): a threshold shift, correctable by normalization. The irreducible disagreement is between the two families of similar severity (16 and 13 severe, intersection 5), which disagree about which episodes. Blaming the outlier would have “fixed” the wrong family

Appendix F — Correction history

This appendix exists because of an adversarial-review finding about the form of the document, not about a number: the narrative of each correction was in the body, interrupting the argument to report the diff. Measured before touching anything with `measurement/densidade-de-avisos.py`: 87 markers distributed over 77 of the 288 paragraphs, 26.7% marked, with retraction being the largest category of repetition. That pre-cleanup output was not preserved: `out/WARNING-DENSITY-2026-08-30.json` was overwritten by the post-cleanup run (112 markers; 102 of 344 paragraphs, 29.7%; body 91 of 297, 30.6%), so the 87 / 77 / 288 / 26.7% have no artifact. Caveat. The two units are distinct and the earlier wording juxtaposed them (“87 markers in 288 paragraphs — 26.7%”), which gives 30.2% if read as a ratio. The number comparable to the one below is that of marked paragraphs.

Correction. The cleanup was partial, and the number after the change has to be here, otherwise this appendix claims an effect it does not demonstrate. In the body (before the appendices, which is what a reviewer reads first), 91 of 297 paragraphs were marked (30.6%) on 2026-08-30, on the 1,914-line version (`out/WARNING-DENSITY-2026-08-30.json`). The 2026-09-21 corrections added marked paragraphs: recomputed on 2026-10-03 with the same script over the 2,058-line text (`_sprint-2026-10-04/A-aging/WARNING-DENSITY-recomputed-2026-10-03.json`), the body has 100 of 307 marked (32.6%), and the whole document 119 of 363 (32.8%), above the 26.7% measured before the cleanup. (Both counts are of the Portuguese text; the English translation needs its own measurement.) The aggregate count is no use for judging: moving material to an appendix does not change the total, and the appendix brings warnings of its own.

Caveat. And the aggregate hides what decides the reading: the warnings are concentrated. Measured on 2026-08-30, §4.2 had 12 markers in 21 paragraphs and §4.1.1 had 5 in 8, while all of §5 stayed between 8% and 17% (per-section counts, not preserved; the artifact holds totals only). A reader who learns to skip the caveat marker learns it in §4.2, and that is where the retractions that change what the result means are. The intervention of 2026-08-30 was therefore surgical and not global, on two fronts: (i) the retractions of §4.1, §4.2 and §5.7.2 came to F-3, leaving a cross-reference; (ii) three paragraphs that are the result (the unusable standard error, the confounder that prevents the strong conclusion, the two properties of the corpus composition) lost the marker without losing a word. The marker there announced as optional exactly what is not.

Caveat. What remains undecided is the third kind: the warnings of *provenance* and of *scope*, which `measurement/densidade-de-avisos.py` promises in its own header to classify and does not classify; the script counts repetition, which is what it measures without judgment. As long as the classification is manual, the remaining reduction is editorial.

The material serves the executable diagnostic (contribution (iv)), and so was not discarded. But it belongs here, and in the body only the corrected state remains, with a cross-reference where the reader needs to know *why* the text is the way it is.

Caveat. What did not come here: validity conditions and scope limits. Those qualify the claim at the point where it is made, and moving them would repeat the error that Figure 1 itself taught: a caveat displaced from the point of reading protects no one.

F-1 — The slots / distinct ratio, withdrawn as a test

Caveat. An earlier version of this section carried here a table of “opposing predictions” (slots/distinct ratio ≈ 1 under a capacity bottleneck versus $\gg 1$ under a policy bottleneck, with 325 printed, though the locked numbers give $583,763/1,787 \approx 327$), presented as the test that separated the two hypotheses. It was withdrawn, for the reasons below. First, the “ ≈ 1 ” prediction is not derived from the capacity hypothesis: it is the hypothesis rewritten in the unit of the ratio, so that observing $\gg 1$ *is* observing slack (a measurement with two labeled regions, not a test with an error rate). Second, the hypothesis it overturned was already dead by the arithmetic above, while the hypothesis a defender would sustain (demand per session above 10) is the one we declare out of scope. Third, and decisive: any concentrating policy produces $\gg 1$, including a correct one. In a corpus with thousands of session fragments and 3,231 chunks of type **daily**, a ratio near 1 would mean serving obsolete digests; it would be the *worse* policy. The ratio measured concentration, and the inference to *defect* came for free.

Caveat. An earlier version of this section sustained the same conclusion through the slots / distinct ratio (325 as printed then; ≈ 327 with the locked numbers), presented as a test that separated capacity from policy. §4.1.1 withdraws that argument and the withdrawal holds here too: any concentrating policy produces a ratio $\gg 1$, including a correct one, so that the ratio measured concentration and the inference to *defect* came for free.

F-2 — Where else the correction did not reach

Each of these was found by adversarial review after the corresponding correction had already been applied at another point in the text. That is the pattern of this class: correcting where the defect was found does not correct it where it also is.

correction applied	where it did not reach	who found it
the importance floor belongs to the channel, but passing the importance floor $\neq$ being eligible	the channel applies three conditions; the importance floor alone gives 13,388 and the three give 108	DeepSeek
the importance floor belongs to the coverage channel, not to the system	the phrase “that the system marked as relevant”, twenty lines below, in the same section	GLM
the 13-type Pearson is fragile to the filter	the partials (-0.709 , $r=-0.843$) that are compared against it	GLM
brief $\neq$ search 583,973 $\rightarrow$ 583,763	Contributions (i), abstract, §4.2, §8.2 none — the value was in five places and all were corrected, but no guard existed	Grok Codex
the equal distribution is impossible “because only 201 fit”	46,295 fit — the measured value was treated as a ceiling, which makes the curve inevitable by construction	internal
the coverage channel window is 7 days	there are two sub-pools, 7 and 30 days; the window of one was applied to the batch of the other	registered prediction
“fossil” $\rightarrow$ “determined” (tone neutralization)	the swap hardened the claim: “determined” is causally stronger than the metaphor, and demanded a counterfactual that did not exist	DeepSeek
the w = 2 column labeled (served)	the defense — “means the dose in force in the <i>shadow</i> ” — lived in the paragraph above the table, and nobody reads a table together with the previous paragraph	DeepSeek
“Nobody measures what the agent actually receives”	a universal over the entire literature sustained by one vocabulary census of a survey of 218 papers	DeepSeek
“the top of the brief is determined by traffic”	the counterfactual measures three chunks; “the top” generalizes to the ten positions	DeepSeek

correction applied	where it did not reach	who found it
the 83.78% attributed to a system decision	9,755 of the 10,899 records are search-counter records, not brief deliveries (the counter marks tracked-search candidacy, not who initiated the call; §3.1, rc6); the caveat existed, but behind a caveat marker in the middle of the paragraph	DeepSeek

F-3 — Six retractions that lived in the middle of the result

These six were inside §4.1, §4.2 and §5.7.2, each interrupting the sentence that carried the finding to report what an earlier version of that same sentence said. The outside reader never saw the earlier version; for them the paragraph alternated between a result and the erratum of a document that does not exist. The corrected state stayed in the body, with a cross-reference; the narrative of the correction is this.

F-3.1 · §5.7.2 — “not measurable” conceded more ignorance than the facts required. The section stated for two days that the third axis of the ceiling was not measurable. There are two distinct things: the *level* of the ceiling under probe exclusion is not recoverable (the corpus e20260826T060003Z.db was an epoch snapshot and was rotated), but the *sensitivity* is (the log of the 350 states is .ndjson and survived). Declaring ignorance is cheaper than measuring, and that is why it is the path of inertia.

F-3.2 · §4.1 — the table closed over two universes. The earlier version listed the historical union next to the live complement; whoever subtracted 67,187 – 11,051 got 56,136 instead of 56,288. Each number was right; the table was not. The row for the 152 chunks served-and-then-deleted is what makes the bridge, and it is now explicit.

F-3.3 · §4.1 — “verbatim from brief.ts:642” for values that are in another file. The *form* of the predicate is there; the *values* 0.7 / 0.7 are in brief-diversity.ts:59–60 and are defaults overridable by environment variable. Presenting runtime configuration as a code constant is what makes a correct number age into a wrong one without warning; the same class that §6 catalogs.

F-3.4 · §4.1 — the mean length was that of another population. The sentence names the 8,928 distilled chunks that pass the importance floor, and the earlier version cited 205 characters, which is the mean of all 14,456 distilled in the corpus, not that of the 8,928. Right number, wrong population, from an *ad hoc* query that left no artifact. The correct value is 232, and it only appeared on recomputing; it survived three reviews because none of them could recompute it.

F-3.5 · §4.2 — “the test separated curation from size”. It does not, and nothing in the section does: there is no variable in the corpus that measures curation. What the tests separate is *filter artifact* from *signal*. The two explanations remain confounded by construction, and the text and the title now say so.

F-3.6 · §4.2 — orphan partials of the number the section itself had withdrawn. The partials (–0.709, *r* = –0.843) appeared without saying over which set they were computed, immediately after the text demoted –0.728 for fragility to the filter. They are the 13 filtered types, and against the Pearson of the 15 (–0.334) “almost unchanged” would be double. Found by GLM (see F-2).

F-4 — 2026-10-04: a verified adversarial review of v1.1

Two voices delivered over v1.1 (DeepSeek and Kimi; three others did not deliver and are not counted). Of the 28 distinct findings, 23 were confirmed against the text and the cited artifacts (_sprint-2026-10-04/REVIEW-A-2026-10-04.md) and are applied here. What changed:

- §9 still said that “the top of the brief is determined” by old traffic after §4.3.2 had scoped the claim to three chunks; it now says the positions of those three.
- §2 said §4.3.1 shows five days with no eligible candidate. §4.3.1 shows 108 eligible on every measured day; the five days of §1 are days without new items.

- The Abstract, Contributions (ii), the §4.3.2 table and §9 now carry the qualifier that the 108-chunk pool was measured during defective session ingestion; “there is never” became “on every measured day there was no”.
- §9: the code yields the existence of the ceiling; its value, 4.86%, was measured by replay.
- §1 and §4.3.1: “the same 33 on every measurable day from 2026-08-21 to 2026-09-21” held for the count, not the set. The set is identical from 2026-08-24 to 2026-09-19, 2026-08-23 has 34, and the month’s 37 is now reconciled (33 + 285042 + three ids of 2026-09-21).
- §4.3.1: the served coverage set stayed the same 108 ids through 2026-09-19 although 285 per-agent chunks were eligible after the repair; the text now says so, and that the reason is not established. The computation, first done by hand during the review, is preserved (`A-rc2/COVERAGE-SET-FROM-LOG-2026-10-04.json`).
- Abstract and §4.1.1 set the brief’s historical 1,787 beside live counts; they now give 1,635 live (1,787 historical). The §4.1 table labelled the search count 9,755 historical; it is live.
- §4.3 attributed “*always with 10 items*” to the current §2, which no longer says it.
- §5.7.2 carried a dangling cross-reference ($\leftrightarrow$ H-x); removed.
- §6.1: “~7,500 per day” had no source. The rate is now 6,614/day (46,295 slots / 7, the closed week), and the three-week extrapolation goes from ~750 thousand to about 722 thousand. The 591,323 of 2026-08-30 is marked as having no artifact.
- §6.1: “those ten” introduced a table of fifteen values; ten were unguarded and five weakly guarded, and the five are now marked.
- §1: `memory` = 1,169 had no generating path in `survey-string-count.py`; replaced by 1,208, the count its positive control computes over the whole text.
- §5.6: “6 probes” against 5 in every preserved probe list; the first run was not preserved, and its count is now declared uncheckable.
- §5.5: “governed by capacity” contradicted §4.3.1; now “by eligibility”.
- §4.2: “`lesson` is at 100% because it has 53 rows” was causal in a section that declares size and curation confounded; now descriptive. “8 are above 32.5%” became 7 above and 1 at 32.5%.
- The status block and §3.2 rule 2 stated the artifact rule as absolute; they now name the numbers the text itself flags as having no artifact.
- §4.3.1: “99.95% admitted to the ranking” now says what admitted means (no eligibility predicate: all 67,187 enter the pre-rank), and “surfaces” became “channels”.
- §4.3.2: an `age` column (125 d for all three chunks) was never defined and no artifact gives it; dropped.
- Open items, item 5: its quotation of §9 now matches §9 (“*nothing yet has been*”).
- Form: the arrows of the Appendix A table became words; the appendices were relabelled into letter order with no gaps (this appendix, formerly H, is now F, after the defect catalog, E), with every cross-reference updated; the work list became the final, unnumbered *Open items*; each bare “the floor” now names which floor; “designated items” is used throughout.

Still open from that review: the [TODO at `deposit`] of Appendix D, which only the deposit closes.

F-5 — 2026-10-05: two verified reviews of rc3, applied in rc4

Two sources. The first is the verification of the rc3 review (Grok and Gemini; 14 findings confirmed, `_sprint-2026-10-04/REVIEW-A-rc3-2026-10-04.md`, cited below as R1 to R14). The second is a further round by Codex, DeepSeek, GLM and Kimi, each finding checked against the text, the code and the cited artifacts on 2026-10-05; the finding-by-finding record is `_sprint-2026-10-04/APPLY-A-rc4.md`. Three quantities were recomputed rather than relabelled, each with a dated artifact: the salience scripts import the production function `calculateSalience`, and the bonus script reads the production constants from the deposited source. What changed:

- Access counterfactual, §4.3.2 and Abstract (Codex-2, superseding R8). The published 2/3/5 $\rightarrow$ 131/129/128 came from `contrafactual-do-topo.py`, whose recency (linear over 365 days, from `source_date`) and importance default (0.5) are not those of `calculateSalience`. Recomputed with the production function (`out/SALIENCE-COUNTERFACTUAL-PROD-2026-10-05.json`, which reproduces the old numbers as an anchor): positions 1/3/4, falling to a tie at 44–46 of 149 (23–47 across instants

- and access-timing assumptions), 54–56 of 201, 40–42 of 144 and 50–52 of 196. “Beyond rank 100” and “exactly 52 places lower” are withdrawn; “128 other chunks” became 22 to 44. The claim that zeroing the access component takes the three out of the top-10 stands.
- Recency mechanism, §4.3.2 (Codex-3). Recency has no floor; the three never decay because their `retention_days` is NULL, the code’s never-decay marker, which 39,130 chunks carry.
 - Tie-break exposure, §5.7.1 (R4). Recounted on the comparator’s key, `calculateSaliency`, instead of the SQL pre-rank expression (`out/TIEBREAK-EXPOSURE-PROD-2026-10-05.json`): 34–42 pairs at second resolution (0.59–0.73%) and 1,197 at day resolution, against 45–59 and 1,656 on the pre-rank expression at the same instants. The 68 / 269 / 917 / 1,656 of 2026-08-29 stay cited as the pre-rank count.
 - Bonus against step, §4.4 and §5.4 (Codex-14, and 14b in §4.4). The 0.0946 and $1.79\times$ used a 0.5 severity multiplier for S1 that only the replay’s summary applies; production applies 0.25. Recomputed: 0.0473 , $0.90\times$ the largest adjacent gap (`out/BONUS-VS-STEP-2026-10-05.json`). The claim that the items crossed several positions is withdrawn, and so is “step $\leq$ distance”.
 - What the exposure count is exact about, §3.1, §4.1, §4.1.1, Abstract (Codex-1). A positive `access_count` marks a top candidate of some tracked search sub-query, a superset of what tracked search returned. The 56,288 is exact as a count of its predicate and a lower bound on non-delivery by the brief and tracked search; the 9,755 is an upper bound on what tracked search returned; the union is no bound in either direction.
 - Derivation premises, §5.3 and §5.4 (Codex-4, Codex-5). The prefix premise is stated, and where it fails (interleaved sub-pools, dedup between coverage candidates) is named. The “if and only if” entry condition holds only when `d` is the only bonused item of its stratum, and then with equality admitting `d` when the tie-break places it ahead of `c_K` (rc5); `b*` as a formula is withdrawn, and saturation rests on proportional bonuses over finitely many states. The generality claims (Abstract, §5, §7, §9) now say “that serves a prefix of that order”; §5.5’s table no longer cites `b*`.
 - Smaller corrections. §1, §4.3.2, §9 and the Abstract: the ceiling carries its regime (R3), and the coverage channel “responds to score only within `last_served` ties” instead of “does not respond” (R7). §9 uses 1,635 live (R1). §5.3: the 17/350 comes entirely from one-second ties (R2). §2: a drafting instruction removed (R6). §4.1: the cohort paragraph is descriptive, not a censoring correction (R9, Codex#7). §4.3.1: the 108 ids are not contiguous (R10); the pool changes the tie structure, not the denominator (R11); “every measurable day” (R14); 2026-09-02 has no briefs (Kimi-8). §1 and Open items: 5,376 is a 672-brief day (R12). §6 and Appendix C agree on where 0.874 is recorded (R13). §4.4: the saturation spread is at least $200\times$, not $220\times$ (Codex#6); the ceiling’s definition carries both conditions of §5.3 (Kimi-10). §5.7.1: $2/9 \approx 22\%$, not 11% (Codex#8). §4.5: the cron windows are named and the residual is not attributed to agents (Codex#9). §7 and Appendix A: which windows are pre-treatment (GLM-4, Codex#10). §4.2: the per-type split by surface is possible and not reported (Codex#11). §6: the attribution method matches §4.3.1 (Codex#12); §6 serves contribution (iv), and its heading changed (Codex#15). §8.2: monotone but responsive only within ties (Codex#13). §4.1.1 and Abstract: 99.98% is an expected coverage (Codex#16). §9 and F-1: 325 is not reproducible; the locked numbers give about 327 (DeepSeek#1). The 0.16% is cut by the whole eligibility predicate, not by path patterns alone (GLM-2). §5.7.2: 104 s, not 98 (GLM-3). Title note: “on every measured day” (GLM-5). Appendix A: the 1,056 s precede the round, not the 20:28Z closure (GLM-6). §5.5: capacity moves the cut (GLM-7). §5.6 and §6: what is module-private (Kimi-4). Appendix A: the $S2 \rightarrow \geq S1$ migration is in Paper B, not in the deviations log (Kimi-6). §5.3: the full exit condition of the fill loop (Kimi-11). §3.3: `saliency.ts` is named (Kimi-13a).
 - Provenance. Status block, §4.3.1, §5.7 and Appendix F: the numbers without an artifact are flagged (R5, Kimi-1, Kimi-3). The last-access dates of §4.3.2 now have one. Appendix D lists the 2026-10-04 and 2026-10-05 artifacts and where `claims_check.py` lives (GLM-1), and where the §4.2 partials are printed (Kimi-2).
 - Independent check of rc4 (`_sprint-2026-10-04/CHECK-A-rc4-B-rc4.md`, D-A1 to D-A3). §4.3.2: the scope-pool sentence mixed two variants; each now carries its own numbers (`prod_acc0`: 76–79 candidates from 43–46 files strictly above; `prod_noacc`: 57–79 from 24–46), and the positions 76–119 are named as stable-sort order inside a tie at 0.69, not a ranking. §4.3.2: a sentence-initial capital restored. §5.7.1: what the remaining ties share is now carried by an artifact, `out/TIEBREAK-EXPOSURE-PROD-`

SHARED-2026-10-05.json (the same script with `--shared-plus 1`, written to a new file; the dated artifact cited before is unchanged).

Not applied here, because this round edits only the manuscript: the same $0.0946 / 1.79\times$ in `DEVIATIONS-FOR-PAPER.md`, `PROTOCOL-CALIBRATION-2026-08-27.md` and `REPLAY-OPORTUNIDADE-2026-08-27.md`, the hard-coded S1 multiplier in `replay-oportunidade.mjs`, the “98 segundos” comment in `claims_check.py`, and writing the §4.2 partials into `out/size-robustness-2026-08-30.json`. The [TODO at deposit] of Appendix D stays open: no new version has been deposited.

Addendum, rc5 (2026-10-05): a regression review of rc4. Codex reviewed the text changed in rc4 and returned ten findings; each was checked against the text, the deposited `serving-*.ts` and the cited artifacts, all ten were confirmed, and the record is `_sprint-2026-10-04/apply-a-rc5.md`. No number was recomputed. What changed:

- Ceiling, §4.4 and §5.3 consequence 1. The stratum condition is a necessary condition under the premises of §5.3, so its fraction bounds the alterable briefs from above; the ceiling is the replay’s measured fraction of states that change at a saturating dose, deduplication included. This replaces the definition attributed to Kimi-10 above.
- Abstract. “For the most part initiated by the agent” claimed more than an upper bound can carry; it now says that most recorded exposure is not a delivery the system decided, and that how much search actually delivered is not established.
- §3.1, §1, §4.1 and the Codex-1 entry above. Search tracking can be switched off per call (`trackAccess = false`, healthchecks and the canary), so the bounds are on tracked search.
- §4.3.2, scope pool. “At most 5 slots” was the quota-pass allowance for agent-specific requests, not a maximum (pinning precedes the quota pass and backfill follows it, rc6); dedup may also reject the three themselves, so pool positions do not establish final membership.
- §4.3.2, positions 23–47. A sensitivity range over a fixed retrospective population, not “the true position”.
- Appendix A. The constant $33 + 108$ set holds from 2026-08-24 to 2026-09-19, inside checks that span 2026-08-21 to 2026-09-21 (`A-rc2/coverage-set-from-log-2026-10-04.json`).
- §5.5 and §9. “Immune” and “deaf to the score” contradicted the corrected response claim; the coverage channel responds within `last_served` ties up to the ceiling.
- §5.4 and the Codex-4/Codex-5 entry above. With the tie-break included, equality can admit `d`; the strict inequality is sufficient, not necessary.
- This appendix. The entry on the access counterfactual had the direction reversed (it is zeroing access that takes the three out of the top-10), and the bonus script reads production constants from source rather than importing a production function.

Addendum, rc6 (2026-10-05): four rc4 corrections that were applied only where they were quoted. A final Codex pass over rc5 confirmed six of the ten rc4 corrections and found that four held only at the quoted place. Each is now fixed as a class: the whole manuscript was searched for it, every hit was decided, and the search tables are in `_sprint-2026-10-04/apply-a-rc6.md`. No number was recomputed or changed.

- Who initiated a search, and what search delivered (Codex-R2). `access_count` is incremented for the top candidates of every tracked sub-query; tracking is the default, and the source comment records that the semantic canary, an automated caller, incremented it until it was switched to untracked (`search.ts:377–396`). The counter records candidacy in a tracked call, not who initiated the call and not whether the chunk was returned. The Abstract, §1, §3.1, §4.1.1, §4.2, §4.3.2, §7, §8.2, §9 and the F-2 table no longer say that the counter shows agent-initiated exposure or that search accounts for most of the exposure. This supersedes the rc5 wording quoted above (“most recorded exposure is not a delivery the system decided”).
- Scope of non-delivery (Codex-R3). Every statement that the 83.78% was never exposed by either surface, never reached the agent or was never shown now carries its scope: those live chunks have neither a brief-log record nor a positive search counter, which is a lower bound on non-delivery by the brief and by tracked search, and non-delivery across all agent-facing search is not established (Abstract, §1, contributions, §2, §3.1, §4.1 heading, table and text, §4.1.1, §4.5, §8.1, §9). Later uses of “never exposed” as the term defined in §2 are left as they are.

- Pinning order (Codex-R4). Pinning is phase 0 of `pickDedup` and precedes the quota pass and the backfill (`brief.ts:438-480`). §4.3.2 said the quota pass came “before pinning” and the rc5 entry above said “pinning and backfill follow”; both now put pinning first, and the §4.3.1 filter list says that F7 runs first although it is listed last.
- Relevance and eligibility (Codex-R7). The coverage channel’s daily reach is set by eligibility, which fixes the candidate population and which the channel exhausted in the measured regime; per-brief selection is where salience acts, within `last_served` ties, saturating at 17/350 in the replay. The categorical forms (“nothing to do with relevance”, “not governed by relevance”, “the score does not decide”, “deafness to the score”, “not by relevance”, “does not predict exposure”) are replaced in §1, §4.2, §4.3.1, §4.3.2 and §5.5, and the Abstract, §4.3.2 and §9 summaries now name both quantities.

Addendum, rc7 (2026-10-05): four rc6 phrasings that kept an exclusive or unscoped reading.

A Codex pass over rc6 found four medium findings. Each was checked against the code or against §3.1 before it was applied, and each was then fixed as a class: the whole manuscript was searched for it, every hit was decided, and the search table is in `_sprint-2026-10-04/APPLY-A-rc7.md`. No number was recomputed or changed.

- Scope of the §4.1 finding (Codex-R3). §4.1 said that “the surface does not show” the fragments that pass the importance floor. It now says that those fragments have neither a brief-log record nor a positive search counter, a lower bound on non-delivery by the brief and tracked search.
- The historical union (Codex-R2/R3). §4.1.1 no longer says that the union “counts what was exposed at some point”. The union combines historical brief-log membership, the chunks deleted afterwards included, with positive search counters on live chunks, and it does not count delivered exposure; the live corpus is the population over which the record predicate is measured, not the one “of which one can say never exposed”.
- Exclusive attribution to the system (Codex-R2). The Abstract (twice), §1 and §4.1.1 (twice) no longer say that only the brief is decided by the system. The brief proactively selects content for delivery; the search counter records candidacy in tracked calls without identifying their initiators, automated callers included, so the counter supports no exclusive attribution.
- What eligibility is (Codex-R7). Eligibility is not a set of paths: the predicate combines path patterns, the importance/pain floor and the age windows, one for the agent sub-pool and one for the global sub-pool (`brief.ts:638-647`). §4.3.1 and §5.5 now say so, and say that the within-tie response holds with eligibility fixed. The Abstract, §4.3.2 and §9 summaries now say that the channel’s daily reach is bounded by its eligible population, which it exhausted in the measured regime. This supersedes the rc6 entry above where it says that eligibility “fixes the candidate population” without naming its components and that per-brief selection “is where salience acts”.

Addendum, rc8 (2026-10-05): two §1 sentences that the rc7 class search missed. A further Codex pass over rc7 (receipt `.remember/adversary-receipt-codex-2026-10-05T100118-80790.txt`, exit 0) proposed two §1 sentences, and both are taken verbatim. The first replaces “exhausts a path-and-age eligibility predicate”, which named two of the three components of eligibility, with a pool “selected by path patterns, the importance/pain floor and age windows”, exhausted in the measured regime. The second replaces “fails, for two reasons, neither of them a shortage of relevance”, a categorical exclusion of the kind the rc6 entry above retired, with “two distinct constraints”: eligibility bounds daily reach, and with eligibility fixed an additive salience bonus changes per-brief selection only within `last_served` ties. Both phrasings are added to the class check. No number was recomputed or changed.

Open items

What is missing, in order of what blocks what. Items are struck through when closed.

Done on 2026-08-28, unless indicated:

Done. §1, §2, §8, §9 written · §5 on 2026-08-27 · Abstract written. **Done.** Figures 1, 2 and 3: each derived from a locked artifact, with its own guard. **Done.** survey count recomputed over the PDF pinned by sha256, with positive control. **Done.** regime break of 2026-08-21–22: explained (§4.3.1), with a dated prediction for

2026-08-29. **Done.** S2, decided by measurement: substantive items at $\geq$ S1 (κ 0.87–0.93), the S1/S2 split becomes an instrument finding (κ 0.31–0.53). Deviation declared in Appendix A. **Done.** defect catalog: 8 in the body (those that changed a reported number), 9 in Appendix E. No separate methods paper.

Missing:

~~1. architecture figure of §2~~ → Done on 2026-08-28: Figure 0, `out/fig0-arquitetura.svg`. It is the only figure that does not derive from an artifact, because it describes structure and not data; ~~2. verify the prediction of 2026-08-29~~ → Done on 2026-08-29, and the prediction was REFUTED: the gate came out `exit 1`, §4.3.1 was rewritten around what the investigation found (two sub-pools; eligible pool of 108 out of 67,187 = 0.161%, exhausted 100% every day), and the new explanation was validated by a differential test `freshSlots=2` vs `freshSlots=0`. Resolved by replacement, not by confirmation;

Caveat. The two items above went three weeks without being struck through, and on 2026-09-21 this list sent an entire session in the wrong direction: reading a worklist from 2026-08-28 and concluding about today’s state. A to-do list that is not struck through is a ruler that ages. What is actually missing:

~~3. adversarial review pass~~ → Done on 2026-09-21: Kimi (Moonshot), receipt `exit: 0`, 682 s, 76 KB over the 1,937 lines. Findings applied in full; see the correction history and `REVISAO-ADVERSARIAL-2026-09-21.md`. Codex ran in the same batch and returned an invalid opinion: four citations that do not resolve against this file, including “*there is no section §4.3.1*”; recorded with the evidence, because «the voice found nothing» and «the voice did not read» are not the same thing.

Caveat. One voice is not the five. The adversarial review of this manuscript was done by one family; Paper B took three. If the deposit requires more, this reopens.

~~original pass:~~ voices of distinct families over the entire manuscript; the recorded lesson is that adversarial review and mechanical census catch disjoint classes of defect, and only the second had been done so far; 4. ~~deposit with the manuscript + artifacts~~ → v1.0 was deposited on 2026-08-30, in Portuguese (10.5281/zenodo.22181415, concept DOI 10.5281/zenodo.22181414; Appendix D). Pending: a new version of that Zenodo record, carrying this manuscript and the artifacts at the cited versions, and the version DOI that the [TODO at deposit] in Appendix D awaits. The grouped amendment of the pre-registration makes sense then: a single registration, declaring the deviations and the new result.

5. Partial. Sweep of claims that have aged (opened on 2026-09-21, three instances closed on 2026-09-21 by adversarial review, not by the sweep): §8.3 (“*that did not run*”), §9 (“*nothing yet has been*”) and §4.5 (“*0 rows*”, now dated and marked as not re-verified). Caveat. The systematic sweep remains to be done. Three instances found by external reading are not a census, and the lesson of today is precisely that declaring a sweep open is not the same as doing it.

Open and declared in the text, from the same opinion: ~~the 52 = 52 reconciliation (§4.3.1 vs §4.3.2), the 25 probe slots (§4.3)~~ → Closed 2026-10-04 by identity measurement (`_sprint-2026-10-04/A-recon-52-and-slots.md`, `measurement/sprint-recon-52-e-sondas.py`). The 52 are one `lessons.md` batch, missing identically on 2026-08-20, 08-21 and 08-22; the “2026-08-20: 85 → 33” was 2026-08-21. The 25 slots are the five probes, identified by `brief_id`, which also add 5 probe-only chunks (organic distinct = 196). ~~the POOL-ELEGIVEL that covers one day for a claim about four (§4.3.1)~~ → closed 2026-10-04 by re-measurement from a preserved database, with a positive control (`_sprint-2026-10-04/A-recon-pool-elegivel.md`, `POOL-ELEGIVEL-2026-08-26-to-29.json`). ~~the comparability control of §5.7.2 that validates by count~~ → closed 2026-10-04 at identity level; ~~the [MISSING] DOI in Appendix D~~ → concept DOI cited, the version DOI comes with the deposit; ~~the pointers to DEVIATIONS-FOR-PAPER.md that Appendix D does not list among the deposited artifacts~~ → listed (`_sprint-2026-10-04/A-recon-5.7.2-appD.md`). The artifact census (`measurement/sprint-censo-artefatos-paperA.py`) still reports 1 missing input (`ts-350.txt`) and 10 Appendix D entries absent from the v1.0 deposit. The next deposit must fix both.

6. **Caveat.** ~~The seven serial filters are not disaggregated (opened 2026-09-21, from Codex’s opinion — the only criticism of it that does not depend on a citation). §4.3.1 attributes the non-exposure of the coverage channel to path patterns; the main pool, 8 of the 10 slots, does not receive the same decomposition. The thesis «policy, not capacity» is sustained for the channel and assumed~~

~~for the main one.~~ → Done: the seven serial filters of the main pool are now disaggregated (§4.3.1, sprint 2026-10-04; `measurement/sprint-desagrega-filtros-pool-principal.py`, `_sprint-2026-10-04/A-filters-disaggregation.md`). The reconstruction on the frozen corpus reproduces the served briefs in 100% of the briefs of 2026-08-24 to 2026-09-07, and leave-one-out on it shows that no filter, lifted, widens the daily surface beyond 33 chunks; F3 (proxy pre-rank with `LIMIT 500`) removes 95.7% of the corpus and changes 0 briefs; F1 (scope routing) and F6 (quota split) narrow it to 8. The main pool is policy-bound (a deterministic top-k, no rotation), not capacity-bound (33 distinct chunks, 0.61% of the 5,376 main slots of a 672-brief day). Caveat: the Codex output naming “seven” was not preserved (only its receipt), so the seven are the stages read from `brief.ts`, not necessarily the reviewer’s list.

~~(the old item 5 said:)~~ Caveat. Sweep of claims that have aged (opened on 2026-09-21). Appendix A said that the interventional study “had not begun”, true on 2026-09-01 and false since then; corrected. That is not necessarily the only one: any sentence written in the present tense about the state of the trial needs to be re-read before the deposit. The method is a reading of the sections that talk about the trial (§4.5, Appendix A, Appendix C, §9), not a substring search.